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ABBREVIATIONS

List of abbreviations:

AnMBR	Anaerobic Membrane Bioreactor	M	Materials
BCM	Building Circularity Methodology	MP	Material Passport
BIM	Building Information Modelling	µCHP	Micro combined heat and power
BM	Business Model	N	Need
BW	Bio-waste	NBS	Nature Based Solutions
CCHE	Collaborative Community of Housing Experts	NZEB	Nearly Zero Energy Buildings
CDW	Construction and Demolition Waste	PV	Photovoltaic
CEBO	Circular Economy Business Model	RES	Renewable energies
Ch.	Challenge	St	Stakeholder
COD	Chemical Oxygen Demand	SaaS	Software as a Service
CT	Circularity Tool	SO	Specific Objective

dAD	dry Anaerobic Digestion	SOA	State of the art
D&C	Dissemination & Communication	SS	Suspended Solids
DfD	Design for Disassembly	St	Stakeholder
DMP	Data Management Plan	ST	Solar Thermal
EPBD	Energy Performance Building Directive	TS	Total Solids
EPP	Efficient Processes and Procedures	UWWT	Urban Wastewater Treatment
GHG	GreenHouse Gas	VNR	Video News Release
KPI	Key Performance Indicators	WFD	Waste Framework Directive
LCA	Life Cycle Assessment	WW	Wastewater
LCC	Life Cycle Cost	WWTP	Wastewater treatment plant

1. EXCELLENCE

1.1 Objectives

Housing represents one of the most important and resource consuming sectors, accounting for about **50% of all extracted materials** and **32% of global energy expenditure**, as well as **a third of water consumption** at EU level (EC, 2017). This sector is also responsible for the **generation of about one third of all** produced **waste** and is associated with environmental pressures that arise at different stages of a building's life-cycle (resource extraction, manufacturing of materials, building design, on-site construction, use and maintenance, demolition and/or recycling)[i]. In a lifetime, an average EU citizen generates at least 160 tonnes of construction and demolition waste and this amount is increasing (EC, 2017). Besides, buildings account for **19% of energy-related GHG emissions** (including electricityrelated), approximately 1/3 of black carbon emissions and 18-33% of Fgases (IPPC, 2014). These negative environmental impacts result from the linear 'take, make, dispose' model, the dominant economic model of our time, traditionally adopted by the decisionmaking of main stakeholders in the housing sector(i.e. construction com-

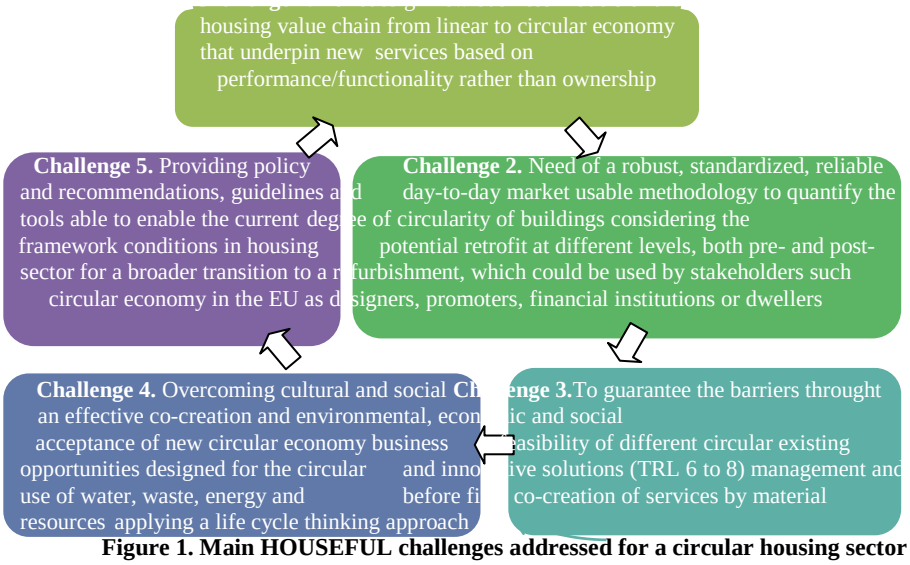

Figure 1. Main HOUSEFUL challenges addressed for a circular housing sector

panies, technological and product suppliers, design teams, municipalities, housing agencies, owners/ dwellers/tenants and policy makers). Besides, buildings lifetime covers a minimum of 40-60 years. This means that the decisions and choices taken by stakeholders severely affect the current and future building's characteristics and its environmental performance from a water, waste, energy, materials and GHG emissions' point of view. Then, not properly planned decisions definitively affect the functionality of the whole utility, related housing practices and lifestyles of dwellers during several generations. The transition from a linear to a circular business economy is being implemented at an industrial scale in Europe. A large number of referenced case studies and innovative projects are focused on the new design of products, smart production in industries and energy efficiency processes. However, this transition has not yet taken place at all levels in the housing sector (energy, water, waste and materials). Most of the time the circularity only focuses on one of these levels (e.g. BAMB project on materials, 4RINEU on energy efficiency and RES, etc).

For that reason, a **new circular thinking approach is required** to enable **better decision-making** on the selection of **circular solutions** at different levels for **all different stages of a building's life-cycle**, promoting the participation and interaction among stakeholders in each stage for an optimal building's functionality and use of resources (water, waste, material and energy) in a co-creation process. Meeting the challenges outlined in Figure 1 requires the redesign of current housing business model, creating new methods for the evaluation of circularity level in housing sector while **co-creating new solutions and services** for its improvement. The transition from linear to a circular business model **in housing sector** will massively contribute to a low-carbon urban economy in future green cities and the

reduction of waste and GHG emissions, contributing to COP21 objectives and the achievement of goals proposed by the 2030 Agenda for Sustainable Development (United Nations, 2015).

The **main goal of the HOUSEFUL** project is to develop and demonstrate an **integrated systemic service (HOUSEFUL Service) composed of 11 circular solutions co-created by stakeholders** in current housing value chain. The HOUSEFUL Service will aim at the circular management and efficient use of **water, waste, energy and material resources** for all stages of European building's life-cycle.

The integrated systemic service will be presented as an **on-line Software as a Service (SaaS),** enabling the **replication of proposed circular solutions offered to stakeholders as Circular Economy Business Opportunities (CEBOs)** at EU level**:** renting, leasing, customer service, capacity building service and/or combination of them. HOUSEFUL will enhance the systemic shift in the housing sector providing the **best strategy** to move towards a **circular economy** while facilitating the **decision-making process** among **stakeholders**.

The co-creation of proposed solutions pursues the following Strategic Objectives for a more circular housing sector:

- The design of new, more **efficient processes and procedures (EPP)** for the construction/refurbishment and demolition phases, leading to a net reduction in the use of resources (materials, energy and water), minimising the waste destined to landfills (reduction from current **40% to 10% in 10 years as realistic scenario**, considering that the aim of circular economy package is 5% until 2030) and selecting/using **sustainable materials** (M) for improving the energy efficiency of buildings *(contributing to the compliance of Raw Materials Initiative, 2008)*
- **The efficient treatment of bio-waste (BW) and wastewater (WW)** through proper use of the building by dwellers. A selection of different **technologies will be offered as new circular services** for the production of treated water for internal **reuse** (i.e. for toilets recharge, irrigation), the generation of **2.600 kWh/y renewable energy** at residential level from biogas for own consumption (i.e. CHP or heat) and the **compost production for gardening** purposes for self and neighbours. In detail, the following technical objectives are foreseen:

- **The recovery of >95% of food waste at home level** by the successful separation of kitchen waste at source, grinding Associated with document Ref. Ares(2018)1767830 - 03/04/2018 and valorisation as **biogas** *(contributing to the compliance of the Waste Framework Directive 2008/98/EC)*;
- **The recycling of >90% of rainwater, greywater and blackwater** for production of **reclaimed water** and **biogas** *(contributing to the compliance of water reuse in Water Framework Directive 2000/60/EC and Urban WW Treatment - UWWT Directive, 91/271/EEC)*
- **High quality biogas production** from grinded BW and blackwater and efficient valorisation **(>90% conversion yield)** as renewable heat and/or power at home level to be able to endure the winter with own heating system *(contributing to the compliance of the Renewable Energy Directive 2009/28/CE)*;
- **High quality compost production** from digestate **(approx. 0.55kg compost/kg digestate)** produced from the joint valorisation of **BW and blackwater;**

- **Reducing the non-renewable primary energy consumption of buildings (E) up to 50%** related to the national regulations requirements (both, for the new and refurbished buildings) by integrating the existing and proposed passive (i.e. energy efficiency and savings by installation of modular facade and roof) and active (from biogas valorisation, etc.) solutions, contributing to the achievement of Nearly Zero Energy Buildings (NZEB) *(contributing to the compliance of the Energy Efficiency Directive and the Energy Performance Building Directive, EPBD 2010)*.

The HOUSEFUL Service will benefit the following stakeholders (St), which will be also involved in the cocreation process of HOUSEFUL solutions:

(St1) **Construction companies, private promoters, technological and product suppliers** by driving them to rethink and redesign their business models towards circular models, while creating competitive advantages for the companies and minimising the environmental impacts associated to their activity. HOUSEFUL will spot and demonstrate the opportunities to offer additional value-added solutions as new services (Table 1).

(St2) **Design teams** including architects, engineers and quantity surveyors will receive guidance on how to assess the circularity level of buildings and the environmental aspects to be addressed in the design stage.

(St3) **Municipalities** and **regional/national housing agencies** will be able to include circular criteria in tender specifications for the construction/refurbishment of public buildings and social housing, resulting in the construction/refurbishment of more sustainable buildings, increasing quality and attractiveness of the buildings' stock and enhancing the quality of life of tenants.

(St4) **Owners/Dwellers/tenants:** improving their understanding of circular buildings by considering the entire life cycle will allow them to make better purchasing decisions, giving them a better idea of how to use and maintain the building in a resource-efficient way and determine optimal end of life strategies. HOUSEFUL will support their decision making process by ensuring transparency on how performance assessments are carried out, and by providing an outlook on the reliability of data that may be used to inform investment decisions and risks.

(St5) **Policy makers:** policy recommendations will be provided to policy makers in order to guide them on how to design a strategy to accelerate the transition towards the circular economy in the housing sector.

HOUSEFUL main objective will be achieved by the consecution of the following **specific objectives (SO)** addressed through planned project activities:

- **SO1**: Developing a robust, reliable and day-to-day market usable methodology to quantify the degree of **circularity of buildings**, at different stages along the whole life cycle of buildings: from design to construction/ refurbishment, operation, maintenance and demolition. Demonstration and validation in **4 selected European buildings**;
- **SO2:** Identifying the key stakeholders and their relationships in terms of influence and interest in housing sector;
- **SO3:** Analysing the specific factors influencing the potential **stakeholders' attitudes and behaviours towards circularity of buildings** and shaping effective engagement towards co-creation; **SO4:** Implementing an innovative engagement strategy to **stakeholders involved in co-creation** activities of proposed 11 solutions as new services by using backcasting exercises. Development of **12 co-creation workshops**, **4 technical workshops, 2 training sessions with stakeholders** and **4 study tours** to the selected European buildings.
- **SO5:** To define the **technical specifications** for demonstration of HOUSEFUL solutions addressed to increase **building circularity** related to water, waste, materials and energy vectors;
- **SO6:** To demonstrate and validate the feasibility of technological solutions at lab scale before demonstration at large scale in 4 selected European buildings;
- **SO7:** To assess the potential environmental and socioeconomic impacts of the circular strategies demonstrated under a Life Cycle Thinking approach using **Life Cycle Assessment (LCA), Life Cycle Cost (LCC) and a social-LCA;**
- **SO8:** To ensure the **marketability** of the HOUSEFUL circular solutions as an integrated systemic service and development of **SaaS for replication** of services in the housing sector;
- **SO9:** To deliver **policy recommendations** at local, regional, national and EU scale to foster the circularity in housing;

- **S10:** To raise the public and stakeholders' awareness by stimulating the acceptance and foster replication of HOUSEFUL services by providing at least **8 independent articles and interviews** around project-related topics, **one Video News Release** (VNRs) for TV distribution and **one web video**, at least **10 press and news releases.** HOUSEFUL project will also be presented in at least **10 events/conferences** related to the topic.

The integrated **HOUSEFUL Service including the proposed HOUSEFUL Solutions**[Associated with document Ref. Ares(2018)1767830 - 03/04/2018] **will be demonstrated at**

large scale in selected **4 representative European residential buildings (Frontrunner buildings)** characterised by different climate and social and legislative/organisational conditions: a building constructed in the 1960's to be refurbished for social housing (Spain), a 1970's built social housing estate to be refurbished (Spain), a centenary building (built 1900) including private and commercial uses (Austria) and a new social housing building constructed in 2017 (Austria).

HOUSEFUL provides new approaches in the development of proposed solutions, combining technological and nontechnological innovative actions, organisational, societal and behavioural innovations to promote leasing and/or renting models of solutions. A key step for their successful development and demonstration is the active participation of stakeholders for their co-creation based on behavioural analysis of stakeholders' intentions. So, engagement activities contributing to a social uptake and impact of proposed solutions in the quality of life, welfare and satisfaction of stakeholders will be performed.

HOUSEFUL solutions are presented as Circular Economy Business Opportunities (CEBOs) and will be offered to stakeholders as **new services at the end of project execution**.

In detail, the following 11 circular solutions will be designed and demonstrated as part of the integrated systemic service:

Table 1. Identified needs, HOUSEFUL solutions and related potential services as new CEBOs for housing sector

Resource	Identified Need (N) – Related Challenge (Ch)	Proposed HOUSEFUL Solution and expected result	Type of service
All *(Material Water Waste Energy)*	**N0.** Online software offering solutions for improvement of building circularity (Ch 1, 2, 4 & 5)	**SaaS:** Software as a Service, aimed at evaluating building circularity while offering **different 11 circular solutions to stakeholders as innovative services** to improve it, achieving an efficient management and use of water, waste, energy and material resources in housing sector.	Software as a Service
	N1. Lack of a representative, standard and feasible methods for the analysis of building circularity (Ch 2)	**Solution 1 (S1):** Development of an innovative methodology for the detailed quantification of building circularity by joining, in a simplest and useable way, the different existing methodologies and tools that partially faces the problem, obtaining and easy-to-use integrated procedure.	Customer service Web-based Tool
	N2. Efficient methodologies for social engagement and cocreation of solutions as new services (Ch 3)	**Solution 2 (S2):** Development of a co-creation methodology based on stakeholder's behaviour and attitudes towards circularity models using backcasting exercises. Demonstration through **12 co-creation workshops**, **4 technical workshops, 2 training sessions with stakeholders** and **4 study tours** to the frontrunner buildings (St1, St2, St3, St4 and St5).	Social innovation service
Material	**N3.** Full knowledge on building's composition for a better resource efficiency, design of new solutions and installation of new objects for improved disassembly and recycling during refurbishment (Ch 1)	**Solution 3 (S3):** Development of **4 Material Passports (MP)** based on advanced **3D model (BIM)** to share asset data across building's value chain. Generation of MP documentation giving an overview of products, components and materials used in the construction object, in sequential years to be updated with a financial valuation. Advise on the architectural possibilities so safeguard as much of the existing materials and value into new building function.	Customer service
	N4. Better use of local materials contributing to more circular buildings (Ch 1)	**Solution 4 (S4):** Sourcing **>50 types** of circular building materials from secondary material platforms or local producers, contributing to debate about quality, origin and potential material reuse at local scale.	Customer/ Capacity building services
Water	**N5.** Efficient treatment and disinfection of rainwater and greywater streams for reuse using green and low-energy consuming solutions (Ch 3&4)	**Solution 5 (S5):** Greywater and blackwater separation. Modifying greywater discharge for treatment using NBS outdoor with energy cost **(<1.5 kWh/m^3)** and commercial O_3/UV disinfection. Obtaining **>95% water reuse** in toilets recharge or irrigation. Rainwater treatment using NBS on facade or roof for **>95% rainwater reuse** in toilets and/or irrigation.	Renting/ leasing of treatment technology
	N6. Efficient treatment and disinfection of un-segregated WW for reuse by green system (Ch 3&4)	**Solution 6 (S6):** Separation of liquid/solid fraction of WW and suspended fraction. Liquid fraction treatment with NBS indoor, at facade and outdoor with a minimum energy cost **(<1.5 kWh/m^3)** and O_3/UV disinfection obtaining **>95% water reuse** in toilets and/or irrigation	Renting/ leasing of treatment technology

Waste	**N7.** Efficient treatment and valorisation of BW and blackwater for biogas production (Ch 3 & 4)	**Solution 7a (S7a):** Joint treatment of blackwater and grinded BW at building (pilot) scale using AnMBR for biogas production, treated water for reuse and digestate for compost. Minimum recovery **>95% food waste** and **>95% organic matter from blackwater** for the potential generation of **>90% treated blackwater for reuse** in toilet or irrigation, **renewable energy from biogas** (linked to S7c) and **compost** (linked to S8) at home level.	Renting/ leasing of treatment technology
		Solution 7b (S7b): Joint treatment of solid fraction of blackwater and grinded BW using dry anaerobic digestion (dAD) for biogas and solid digestate (compost) production. Minimum recovery **>95% food waste** and **>95% organic matter from WW** for production of **renewable energy from biogas** (linked to S8) and **compost** (linked to S7c) at home level.	
		Solution 7 (S7c): Biogas cleaning for *(1)* storing and possible use in win-	
		ter times according to energy needs of buildings or surrounding infrastruc- ture in neighbourhood (potential collaborative economy) and/or *(2)* CHP or heat production and consumption by the own building (**2600kWh/y renewable energy from biogas**) or other co-creation possibilities.	
	N8. Need of high quality fertiliser/compost from local origin (Ch 3&4)	**Solution 8 (S8):** Stabilised BW from anaerobic digestion systems (AnMBR or dAD) for use as compost in local gardening (public parks, training activities in schools, etc.) substituting commercial compost. Use of low-energy consuming and green stabilisation technologies for final production of **0.55 kg compost/kg digestate.**	Renting/ leasing Capacity building service
	N9. Optimal management of waste/resource at the end of building's life cycle (Ch 3&4)	**Solution 9 (S9):** Analysis of maximum recovery and valorisation potential of existing materials in buildings (i.e. envelope, facade, etc) to guarantee the best wastes management (reduction of **30% CDW** disposed in landfills). Selection of safety procedures for disassembly of construction elements.	Customer service
Energy	**N10.** Active and passive energy efficient solutions for buildings based on greener and functional elements (Ch 3,4 & 5)	**Solution 10 (S10):** Design of energy efficient facades and roofs using modular and more sustainable materials selected according to circular criteria. Assessing on the optimisation of global refurbishment actions according to energy efficient criteria. Including possible energy generation from the building.	Customer service Renting/ leasing
	N11. Guarantee energy saving/production in buildings (Ch 3,4 & 5)	**Solution 11 (S11):** Co-creation based on performance rather than ownerwhip for energy systems (i.e. ST, PV) finally installed in each building (Pay per performance business model). Monitoring of energy saving by active and passive measures adopted from material and energy valorisation points of view.	Renting/ leasing Capacity building service

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Proposed **11 solutions** have been **selected from a preliminary analysis of common needs and barriers in housing sector evaluated by HOUSEFUL project partners, which are already stakeholders**. HOUSEFUL proposes innovative solutions as new services to all stakeholders in housing sector in order to increase the circularity of a current linear sector with high environmental impacts in water, waste, materials and energy vectors.

Frontrunner buildings will act as reference for replication activities of proposed solutions. Besides, HOUSEFUL predicts **10 new demo sites**, **named as Follower buildings** that will be identified **during the 1st year** of the project execution and engaged for replication of proposed solution as new services in other EU regions by the execution of planned networking/exploitation and dissemination activities. Additional funding sources will be sought to enhance the effective marketability and replication of proposed solutions as new services in **Frontrunner and Follower buildings**, making possible the replication of HOUSEFUL services by new stakeholders in EU regions after project execution.

It is worth to mention that a total amount of **32 Letter of Interest have been collected** from European public institutions and companies' representatives of housing sector, which will be invited to currently planned project meetings.

1.2 Relation to the work programme

HOUSEFUL responds to the topic "CIRC-01-2016-2017: Systemic, eco-innovative approaches for the circular economy: large-scale demonstration projects [b] Systemic services for the circular economy (2017)".

Systemic services for the circular economy. CIRC-01b-2017	
Specific CHALLENGE of the topic	***What does the HOUSEFUL project do?***

Replace current linear economic models with circular models of production and consumption; reduce GHG emissions.
HOUSEFUL facilitates the transition to circular models in housing sector, improving resource efficiency and reducing the negative environmental impact throughout the full life cycle of residential buildings. We propose a conceptual framework for the transition from a linear to circular Business models (BM) as key elements to reduce GHG emissions.
[…] systemic approach to eco-innovation that encompasses value & supply chains in their entirety and engages all actors involved in such chains […] impacts that transformative innovative solutions can have on economy, environment & society
HOUSEFUL adopts a systemic approach that goes beyond conventional BM by demonstrating an innovative integrated systemic service for residential buildings. The project comprises actors from different areas collaborating in a symbiotic manner including: construction experts (ITeC), Social Housing Organisations (AHC, Neues Leben, CECODHAS), environmental & technological experts (ALCN, AIGUASOL, LEITAT, Cartif, IDP), social architects (GWR), social innovation and humanities areas experts (WE&B), circular BM experts (Turntoo, LGI), IT applications (VISUM) and communication experts (ICE). The project will promote and facilitate the interaction of these actors with different stakeholders of the housing value chain identified from the beginning of project execution (including building developers and investors, design teams including architects and engineers, construction companies and contractors, facilities managers, housing organisations, public bodies and dwellers).
Bring end-users closer to the design and production phases, […] boost new consumption patterns that add greater value and reduce over-production, waste and other negative environmental impacts and involve end-users in circular economic model
Circular economy business models to the housing value chain will increase new demands and open new markets for a more efficient use of resources in the housing sector. HOUSEFUL proposes a path towards an EU reindustrialisation economy, creating better market conditions for products and services. HOUSEFUL will also provide property owner and inhabitants with a tool for the analysis of the energy and resource efficiency of the build-
ings they live in, empowering them to make informed decisions regarding improvements and innovations to be implemented, driving more sustainable consumption patterns, and ultimately placing them in the position of manager, owner and even energy provider, instead of consumer and waste producer.

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SCOPE of the topic	***What does the HOUSEFUL project do?***
Demonstrate through large scale projects the economic and environmental feasibility of circular economic business models that underpin new services based on performance/functionality and/or on mass customisation.	
HOUSEFUL will demonstrate an innovative circular economy BM for the housing sector, looking at the whole lifecycle of a residential building from the selection and extraction of materials, manufacturing of construction products, construction, use and maintenance, demolition and waste & water treatment. The environmental and economic feasibility of new services, as underpinned by circular solutions, will be demonstrated in 4 representative European buildings named as Frontrunner residential buildings. In order to ensure a major potential implementation of HOUSEFUL services and solutions, also theoretical scenarios corresponding to different all over Europe latitudes will be implemented through the analysis of several building archetypes in at least, 6 geoclusters zones. Also, an adaptation of the obtained results and services will be made by replication activities foreseen in 10 Follower buildings to ensure the large-scale impact.	
Solutions should entail the environmentally sustainable recovery, recycling and/or re-use of resources and energy flows, including by cross-sectorial symbiosis from resources to marketed products	
The circular solutions demonstrated in the 4 frontrunner buildings will address the most critical environmental impacts associated with the entire life-cycle of a residential building. New services will be provided promoting efficient processes and procedures in the construction and demolition phases, and systems for the efficient treatment of BW and WW for reuse within a building and the generation of renewable energy for self-consumption.	
Adopt an eco-approach through a Life Cycle Thinking to all forms of innovation which fosters new forms of collaboration between stakeholders; support of co-creation	
The HOUSEFUL business model includes a wide variety of circular solutions, from sustainable technologies to innovative services for the housing value chain, boosting dwellers' choice for sustainable consumption patterns. Environmental, social and economic impacts of the 4 frontrunner buildings will be assessed by applying LCA, LCC and social assessment. New forms of collaboration among stakeholders will be encouraged.	
Business models will require support to end-users in the transition to circular economy by sharing knowledge and activities	

A SaaS will be created to support actors from the housing value chain to redesign their BM according to circular economy principles. A user-friendly circularity tool for professionals but also for owners/dwellers/tenants will also be delivered to raise awareness of circular economy models. A Business Plan of the HOUSEFUL model will be developed to provide circular services/products as new leasing and/or renting conditions to the housing sector. Special attention will be paid to the leasing and/or renting models for such physical solutions developed (i.e.: modular facade, AnMBR, etc.). Policy restrictions and safety guarantees will be reviewed and overcome major risks in the new commercialisation of solutions under new circular business models adapted to services.
Contributions of regional/national schemes under the ESIF, such as under the ERDF, or other relevant funds such as IPA II
HOUSEFUL will collaborate with pre-identified local authorities from two different geographical regions (Austria and Spain) to find EU-complementary funding schemes, like under the RIS3, the ESF, through regional funds, and ethical banks, to cross-finance the renovation process as such and generally deepen the impact of project's results at local level.
Possible regulatory barriers should also be addressed
Drivers and barriers for the circular economy in the built environment will be addressed including regulatory, but also financial, societal and organisational ones. The incorrect, sub-optimal or delayed implementation of the environmental policy framework (e.g. Energy Efficiency Directive) might lead to insufficient savings, having a negative impact on 2030 target and the decarbonisation of the economy in 2050. Thus, HOUSEFUL will identify regulatory barriers and propose recommendations to overcome them in order to facilitate the market up-take of innovative solutions.

1.3 Concept and methodology

1.3.1 Concept

1.3.1.1 Overall concept

The housing sector is a major contributor to current global problems of pollution, resource depletion and climate change, representing from a life cycle perspective one of the most important and resource consuming sectors at EU level: accounting for about 50% of all extracted materials, 40% of final energy consumption, 33% of water consumption and 33% of all produced waste (EC, 2014). Buildings in the EU have a floor area of 25 billion m^2, with about 75% of this being residential buildings. The low annual renovation rate of existing buildings (only 0.4-1.2%) (EC,2016), the low annual demolition rate (only 0.1%) and the low annual construction rate of new buildings cause that much of the EU building stock is in need of renovation, with more than 40% of it built before 1960 and 90% before 1990 (EU Parliament, 2016). In addition, population growth, migration to cities, and increasing levels of wealth and lifestyle changes will all contribute to significant increases in building resources use.

As a consequence of the environmental pressure caused by the housing sector, the EC has recently launched a set of policy initiatives promoting the construction and refurbishment of buildings in a resource efficient way: the *Roadmap to a Resource Efficient Europe- COM(2011)571* highlights buildings as one of three key sectors to be addressed to make significant resource savings; the *Resource Efficiency Opportunities in the Building sector COM(2014)445* describes the need for a common EU approach to assess the environmental performance of buildings throughout their lifecycle, considering relevant resources such as energy, materials, water and waste; *Towards a circular economy: A zero waste programme for Europe-COM(2014)398* encourages the design for better management of CDW; and *Closing the loop - An EU action plan for the Circular Economy-COM(2015)614* considers "construction and demolition" as one of the five priority areas to be tackled in the context of the circular economy, and presented four legislative proposals on waste, containing targets for landfill, reuse and recycling, to be met by 2030. On November 2016, European Commission presented a package of measures known as *Clean Energy for all Europeans* to speed the clean energy transition in Europe. The package includes sets of legislative measures and several communications that will have an impact on housing sector, highlighting the proposals for revising the EPBD 2010, the Energy Efficiency Directive, the Renewable Energy Directive, the Ecodesign Directive and a spe cific communication called *Accelerating Clean Energy in Buildings - COM(2016) 860 Annex I.*

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HOUSEFUL project is aligned to the EU policies and initiatives in terms of combating climate change and sustainable development, supporting their targets and recommendations to enable the transition of the housing sector towards a circular economy. **HOUSEFUL aims at improving resource efficiency** (energy, materials, water and waste) **of residential buildings and bringing them into the circular economy**, supporting EU policies and priorities.

Under traditional business models, investments related to retrofit actions are considered a liability, even contrary to clear evidence that the market value of retrofitted buildings is higher than energy-poor buildings. In addition,

monetary savings redeemed from retrofit actions, specifically in the housing sector, do not yet out-perform the initial investment. To overcome both barriers, various EU projects (See Table 5) have developed new solutions aimed at lowering the investment requirement and maximising monetary savings. Although there are physical and economical limits to these strategies, circular economy principle allows for their implementation and incorporation into economically viable and sustainable solutions and methodologies that convert retrofit investments from a liability into an asset, viable to renting and/or leasing models, thereby fully valorising the retrofit actions.

Stakeholders in housing sector require attractive services and incentives to promote their involvement in valuable actions contributing to a more efficient use of resources.

It is not always possible to close all the circles within one single building (especially if it is a smaller building) and we will anticipate the limits and in where closing bigger circles make more sense (like at district or city level). To tackle all these challenges, HOUSEFUL has gathered industry experts (SMEs and large companies), RTDs and public administration figures who will collaborate closely to achieve and demonstrate the project objectives and the creation of new CEBO with high replicability at EU scale.

HOUSEFUL CONCEPT

In HOUSEFUL, an innovative paradigm shift towards a circular economy for the housing sector is proposed to overcome current Challenges in housing sector (Figure 1) by demonstrating the feasibility of an **integrated systemic service** composed of **11 systemic solutions** in the fields of **energy, materials, water and waste**. The integrated service will be driven and promoted as a **SaaS, aiming at evaluating building circularity while offering different circular solutions as services to increase it for all stages of building life cycle.** HOUSEFUL concept (Figure 2) pursues the transition from linear (take, make, consume and dispose model) to circular business models for the stakeholders by **proposing new CEBOs**. Indeed, HOUSEFUL favours the migration of the housing sector from the current PITO models (Product in, Trash out) to a new DIDO model (data in, data out)**.**

From HOUSEFUL, an **innovative holistic model** will be ready to be offered to stakeholders in housing sector as innovative and integrated services, replicable to other residential buildings. A **Software as a Service (SaaS)** will be developed in order to **improve the replication** of proposed 11 circular solutions as services based on performance/functionality rather than ownership.

The SaaS will offer the following features: 1) a Circularity Tool (CT) to quantify the circularity level of residential buildings. Users will be able to work with data and calculation methods at two defined levels: a first level (free access) with clear and adapted KPIs (such as water reuse ($m^3/m^3 \cdot day$), total primary energy demand ($kWh/m^2 \cdot yr$), share of renewable energies (%), etc.) and outputs and a second level (a pay per use) with really oriented results, where different types of interventions to enhance the circularity of the building will be suggested and quantifying also the effect of these interventions on the building. The CT addresses the current information deficit by proposing, whenever possible, a set of clearly defined and measurable indicators, for the assessment of the environmental performance of residential buildings; **2) an interactive sharing platform** created with the primary aim of promoting **the co-creation of new CEBOs** by stakeholders in housing sector. The platform will contain **successful circular solutions and services**, the ones demonstrated in Frontrunner buildings (11 HOUSEFUL Solutions) but also other successful experiences collected, with the aim to promote the replication of these initiatives and facilitate the collaboration among stakeholders.

The SaaS will help the housing value chain work on their value proposition, clients, activities, partners, cost structure and revenue streams in order to define a successful company. At this moment, there exist some circular products offering repositories or greener building design (see C2C) or new brand methodologies (Level(s) by EC), but not the combination

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of both. **SaaS service will not only evaluate the building circularity** (as Level(s) seems to be), **but will offer explicit solutions to improve it in a service mode**. HOUSEFUL pursues the transition from linear (take, make, consume and dispose model) to circular economy business models for the stakeholders in current housing value chain, where sharing, reusing, repairing, refurbishing and recycling in an almost closed loop are the key principles of the circular economy.

Different systemic solutions offered in HOUSEFUL **will be demonstrated in 4 frontrunner buildings,** including social housing, with the aim to validate their technical, environmental, social, economic and regulatory feasibility.

The **4 frontrunner buildings** are located in Spain and Austria: 1) "El bloc dels Mestres" building, located in Sabadell (Spain) and constructed in the 1960's is going to be retrofitted for social housing; 2) "La Grípia" building, located in Terrassa (Spain) is a social housing building constructed in the 1970's; 3) "Baumgarten" a centenary building located in Vienna (Austria) and 4) "Donaufelderstrasse", a new social housing constructed in 2017 and located in Vienna (Austria). The buildings were selected with regard to: (***a)*** a spatial/geographical distribution which will guarantee the feasibility and replication of solutions, (***b)*** differences in social, cultural and current practices on housing, ***(c)*** differences in national regulations regarding construction and refurbishment, ***(d)*** common in European building archetypes, scale and number of dwellers per building; ***(e)*** climate differences in promoting the different

characteristics and typologies of residential buildings, and ***(f)*** common challenges shared by construction companies, related professionals and regional/national housing agencies. As it can be seen 3 of the 4 buildings are

1 Stuchtey et al, 2016.
2 EUROSTAT, 2017. Electricity and heat statistics
3 Åsa Stenmarck et al, 2016. Estimates of European food waste levels. FUSIONS EU Project. Grant Agreement: 311972.
4 WTE Ltd (2017) Wastewater Produced In The Home. https://www.wte-ltd.co.uk/

Figure 2. HOUSEFUL concept

social housing buildings, since one of the HOUSEFUL challenges is to demonstrate solutions that can be implemented

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for improving the circularity level of low-income buildings. In relation to energy poverty, savings from HOUSEFUL solutions seeks to respond to the increasing difficulty many tenants (more than 50 million people in the EU are at risk of energy poverty (EU Parliament, 2016) experience in paying water, waste and energy bills.

DEMO 1 – Old Building (60's)	
Location: Carrer de Campoamor, 98, 08204 Sabadell (Spain) (Approximately 20 km North of the City of Barcelona). **Basic data**: Constructed 1960s; Ground floor + 8 residential floors, 2 flats/floor; 16 x 100m^2 approx. homes with 4 bedrooms +2 bathrooms; 14 unoccupied, all in need of deep refurbishment.	

Description: The so-called teachers' block (*"El bloc dels Mestres"*) is a residential building built in the early sixties adjacent to an educational centre the current Joan Sallarès i Pla school), and initially intended as a teachers' residence. It has become increasingly disused over recent years. There are now just two teaches left in the block and they are due to leave before the end of 2017. **The AHC have a budget of about 1.5M Euros already committed for the refurbishment** of the project (see letter of commitment in annexes) including the following planned actions: Building envelope refurbishment with insulated façade; Solar thermal; Maintenance of current size and distribution; Use as rented social housing for large families at risk of social exclusion. The timetable is to have the definitive project approved by March-April 2018 and launching the public tender procedurefrom May 2018 (see the details in the Gantt attached in section 3.1.2).
HOUSEFUL will provide an advance in this building circularity by offering solutions for their calculation (S1), offering new methodologies for stakeholder engagement (S2) and co-creation of innovative solutions on materials (S3, S4), water (S5 in a minimum of 8 homes), waste (S7, S8, S9) and energy resources (S10, S11).

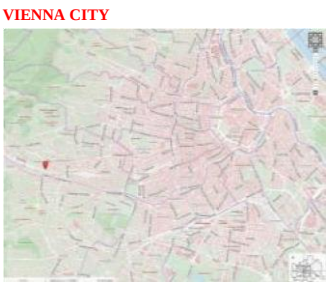
DEMO 2 – Old Building (70's)

Location: C/Tibidabo 1, 3 /Ptge Calaf 81, 83, 85, 08227 Terrassa (approx. 30 km North, northwest of the City of Barcelona).
Basic data: Constructed 1970s; 2 four story buildings; 41 residential units (some 98m^{2,} others 69m^2); 3750m^2 residential area in total; All occupied. All in need of refurbishment using circular solutions promoted in HOUSEFUL.

Description: "The Grípia" is the name given to five blocks of residential buildings built in the 1970s. The AHC owns the buildings and rents the properties as social housing. HOUSEFUL project is seen as a valuable opportunity for the AHC to go further with its demonstrated commitment to sustainable development and to innovate in the Circular Economy. If HOUSEFUL is financed, the development of the executive refurbishment will become a top priority for the AHC (see letter of commitment in annexes). This means that refurbishment could commence within 25 months of the start of HOUSEFUL the project. Co-creation of works for the successful refurbishment will be done in stages to minimise disruption to the tenants who will continue to reside in the buildings throughout the project. Work will be completed within 24 months. A full 12 months for co-creation, monitoring and evaluation will be possible in some of the flats (minimum 2 flats from 1 of 5 blocks). HOUSEFUL will provide an advance in this building circularity by offering solutions for their calculation (S1), offering new methodologies for stakeholder engagement (S2) and co-creation of innovative solutions on materials (S3, S4), water (S5 in a minimum of 2 homes), waste (S9) and energy resources (S10).

DEMO 3 – Centenary building

Location: Vienna, Baumgartenstrasse 93. The building is located in the 14th district of Vienna ("Penzing").
Basic data: Constructed 1900. 3 floors space of 462.6 m^2 (occupation floor space of 370,1m^2) with a chemistry and microbial lab in the basement, an office area on the ground floor, and a residential unit on the first floor and a loft.



Description: DEMO 3 is a residential building, including an office, laboratories and residential space. It is privately owned. Rental agreements are unlimited in time. The building is due to substantial refurbishment. This building reflects a typical building category of Central European centenary buildings that were not destroyed during the war. Main demo targets are to demonstrate the feasibility of HOUSEFUL services co-created by stakeholders and of-

fered to the users (and possibly the neighbours) of the building, while at the same time reducing the energy demand of the building, and closing water and nutrient cycles by recovering water and nutrients from waste water and organic waste. This will be achieved by structural modifications concerning pipe installations, roof structure, facade change, integration of winter gardens, and modifications in the garden. Ma HOUSEFUL solutions will also involve the use of circular materials to almost passive house standards but includi new functionalities on facades and roof such as nature based solutions for WW treatment. The energy intake after t renovation will be done with photovoltaic and burning of biogas that is directly produced in the building. All the organ waste materials that are produced in the building will be used for biogas generation and further in-vessel compostin The WW will be treated in plant-based treatment systems and the dissolved nutrients further used for fertigation in urb farming applications, also in a greenhouse setup. The overall of proposed solutions will provide new opportunities advance in this building circularity, previously developing an innovative method for this calculation (S1). The solutions will be based on new methodologies for stakeholder engagement (S2) and co-creation on materials (S3, S4 water (S6), waste (S7, S8, S9) and energy resources (S10, S11).

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DEMO 4 – New building (2017)	
Location: Vienna, Donaufelder Strasse 115. The building is located in the 21st district of Vienna ("Floridsdorf"). **Basic data**: Constructed 2017. Total area of 4.626 m², with 54 apartments, 1 enterprise, 1 supervised flat-sharing community for young people. Places for collective use exist. Exterior walls made of reinforced concrete.	
Description: DEMO 4 is a residential building with occupations of all 54 flats. Technically, the building corresponds to Austrian standards and is comparable to approximately 70% of residential buildings in Austria. Therefore, the DEMO 4 is an interesting role model for replication and dissemination activities in HOUSEFUL. NEUES LEBEN owns the buildings and rents the properties as social housing. The HOUSEFUL project is seen as a valuable opportunity for NEUES LEBEN to go further with its demonstrated commitment to sustainable development, to observe interventions in the other HOUSEFUL demos and to incorporate circular economy in future projects. Results from the HOUSEFUL calculations and interventions in the demos will be subject of discussions in organised stakeholder processes, introducing the topic of circular economy in the residents' and neighbour's daily life. HOUSEFUL will provide an advance in this building circularity by offering solutions for their calculation (S1), offering new methodologies for stakeholder engagement (S2) and co-creation of innovative solutions on materials (S3, S4), wastes (S9) and energy resources (S11).	

Co-creation among stakeholders: a key step for the successful demonstration of HOUSEFUL solutions

A key aspect of HOUSEFUL project is the co-creation value for achieving understanding and effective engagement among stakeholders in the definition of what the building circularity is and what kind of solutions can be provided to increase its circularity at different levels and stages in building's life cycle. HOUSEFUL will address the notion of "success" of the project in a process of **stakeholder interaction** for **identification (WP2,3)**, analysis (WP3), **development (WP4,5), replication and promotion (WP6,7) of the services** offered to stakeholders from solutions to be demonstrated in 4 Frontrunner buildings. Therefore, activities seeking effective engagement will be a central and integrative element for the project and will be developed based on factors influencing stakeholders' attitudes and behaviours towards circularity in buildings (including tenants and decision makers). Engagement activities includes **12 co-creation workshops** based in backcasting exercises (social sciences)**, 4 technical workshops**, **2 training sessions** with stakeholders and **4 study tours** to the Frontrunner buildings. Effective engagement is going to be achieved by a multidirectional approach based on social sciences (backcasting exercises to identify ideal futures and prepare reliable scenarios with a selected combined group of stakeholders). Social interaction will address needs and concerns expressed by the stakeholders in the behavioural analysis, and integrate them into the creation of services of building circularity through knowledge brokerage practices. Based on this innovative strategy defined for HOUSEFUL, the following circular solutions will be developed and demonstrated at building scale in each Frontrunner building.

Solution 1 (based on **N1**):	**Definition of a new method for the analysis of Building circularity**	Potential type of service:	**Customer service, Web-based tool**
Interested Stakeholders:	St1, St2, St3, St4, St5	Linked building life cycle:	All stages

Solution 1 is focused on the development of an innovative methodology for the quantification of building circularity considering current management and use of water, waste, energy and material resources in all stages of housing value chain. Stakeholders in housing sector (such as businesses and dwellers) have little knowledge about the potential benefits of a circular economy and tend to be reluctant to adopt new circular business models. In HOUSEFUL, a new methodology will be created to calculate the **Circularity Level** of a residential building to aware the housing value chain about negative impacts caused by the use of traditional linear business models. HOUSEFUL building circularity methodology
(BCM) will be developed **in WP2** and will use these existing references by joining them in the appropriate way, and

improve them by adding thumb rules and tricks to easily and robustly define the degree of circularity of a building.
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Building circularity will be calculated for Frontrunner buildings based on the calculation procedures and KPIs defined in HOUSEFUL project under a life cycle thinking approach**.** This solution will be offered to stakeholders as a web-based tool for the evaluation of European buildings' circularity. The web tool, which will also introduce specific solutions and circular services, will be improved in a loop process and will be the basis of the SaaS developed by VISUM **in WP6**.

Frontrunner building:	Demo 1 - (60's)	Demo 2 - (70's)	Demo 3 - (1900)	Demo 4 -(2017)
	☒	☒	☒	☒

Solution 2 (based on **N2**):	**Social engagement for co-creation**	Potential type of service:	**Capacity building service**
Interested Stakeholders:	St1, St2, St3, St4, St5	Linked building life cycle:	All stages

A cross-cutting solution, offered to stakeholders as a service (covering energy, materials, water and waste issues) will be provided in HOUSEFUL consisting in offering capacity development activities for co-creation of new circular business opportunities to stakeholders in current housing value chain. Currently, most of the housing sector businesses are lacking understanding to develop products or services in a circular approach and to uptake solutions. The cocreation process will address the variety of levels of acceptance from the political perspective, the community perspective and particularly, at the individual perspective of households and tenants in terms of the attitude and behavioral response or to the implementation and adoption of the HOUSEFUL solutions. Social innovation activities offered by this service include the analysis of drivers, enablers and barriers driving stakeholder decisions and behaviours. Although same research question is proposed, the empirical approach will differ from the context variation of each demo sites, specifically at individual level of end-users (tenants and householders). In thus, when working in demo-site with potential tenants the approach will be based in the provision of stimulus information materials and observation and integration of changes in attitudes and behaviors. When approaching tenants already leaving in the demo-sites the co-creation process should start earlier to foresee attitudes and behavior towards the potential change.

	☒	☒	☒	☒

Solution 4 (based on **N4**)	**Searching local building material**	Potential type of service:	**Customer service, Capacity building service**
Interested Stakeholders:	St1, St2, St3	Linked building life cycle:	Design, refurbishment, dismantling

	☒	☒	☒	☒

This solution is focused on sourcing >**50 types of building materials** (existing or new commercial products) from secondary material platforms, databases and/or local producers, contributing to debate about quality, origin and

Based on the findings and insights from the previous social and psychological analysis a definition of engagement activities and a knowledge sharing process will be based on online (web-access, online events, etc.) and offline sharing process based on face to face interactions (in a form of workshops). In this sense stakeholders' will take part in the process of knowledge exchange (e.g. successful initiatives related to smart and sustainable buildings) aiming at fostering business and social innovation towards a circular economy. In HOUSEFUL, the completion of project co-creation activities will be led by WE&B in **WP3** and include the stakeholder analysis and the development of **12 co-creation workshops** based in backcasting exercises**, 4 technical workshops**, **2 training sessions** with stakeholders (public bodies, builders, architects, dwellers, etc.), and **4 study tours** to the Frontrunner buildings. Specific **communication materials** to guarantee the stakeholder engagement in co-creation activities will be developed **in WP7** by ICONS such as a call-to-action video (based on animations and infographics), different types of flyers customised for stakeholders in each Frontrunner buildings and other including web-based social media.
Frontrunner building: Demo 1 - (60's) Demo 2 - (70's) Demo 3 - (1900) Demo 4 -(2017)

Solution 3 (based on **N3**): **Material Passport + BIM**		Potential type of service:	**Customer service**
Interested Stakeholders:	St1, St2, St3	Linked building life cycle:	Design, refurbishment, dismantling

It is well known that the construction industry suffers from inefficient practices, resulting in high amounts of waste, duplication of processes and increased costs. A solution focused on sharing data about materials used in new and existing buildings and their impacts on the environment in a digital format to stakeholders across buildings' value chain will be provided. As part of this solution, Building Information Modelling (BIM) and Material Passport (MP) will be jointly provided as services that will be offered to increase productivity and efficiency in the built environment, allowing for better decision making along the value chain, from design to construction/refurbishment, operation, maintenance and demolition. In HOUSEFUL, **MP** based on advance **3D model (BIM)** will be carried out in **each Frontrunner building** to share asset data across building's value chain. Generation of documentation in a MP, developed by TURNTOO **in WP4**, will give an overview of products, components and materials used in the existing construction objects, in sequential years to be updated with a financial valuation. Advise on the architectural possibilities so safeguard as much of the existing materials and value into new building function. BIM will support the development of each MP, and it will be performed by ITEC for all Frontrunner buildings **in WP4**. Frontrunner building: Demo 1 - (60's) Demo 2 - (70's) Demo 3 - (1900) Demo 4 -(2017) potential material reuse at local scale. The search will be focused on EU cAssociated with document Ref. Ares(2018)1767830 - 03/04/2018ertified products including technical and environmental specifications clearly contrasted, allowing the complete material characterisation and calculation of potential building impact along the overall buildings' value chain -manufacturing, design, construction, use and demolition phase. Results from this search will be included in a "Circular materials database". The database itself will not include only certified products (wood, steel, etc.), but also cost and supply conditions linked to these products. Some of these (more circular) products will be widely defined and detailed for potential use in Frontrunner buildings such as those contributing to the improvement of energy efficiency or other, like paints, contributing to the improvement of indoor and outdoor air quality. In HOUSEFUL, the analysis of potential use of certified building material will be essential for the improvement of building circularity. This will also facilitate the potential modelling of building characteristics and the successful KPI quantification for impact analysis. Results can provide new circular business opportunities at local scale. This solution will be led by ITEC and performed **in WP4**. Frontrunner building: Demo 1 - (60's) Demo 2 - (70's) Demo 3 - (1900) Demo 4 - (2017)

Solution 5 (based on **N5**)	**Efficient treatment and reuse of rainwater and greywater**	Potential type of service:	**Renting/ leasing**
Interested Stakeholders:	St1, St2, St3, St4	Linked building	Design, refurbishment and

		Linked building life cycle:	Design, refurbishment and use

Greywater and blackwater can be separated with retrofitting measures in existing buildings. Greywater can be treated using innovative NBS for indoor application in multi-level green walls with **minimum energy cost** (**<1.5kWh/m^3**) and disinfected using commercial O_3/UV systems for **>90% water reuse** (we assume <10% losses by evapo-transpiration). Water meters at household and building level can show the amount of water reuse for toilets recharge or irrigation. On the other hand, rainwater can be also pre-treated with NBS on the façade (developed in EeB-2016 project by ALCN) or roof area (developed in an Austrian project by ALCN in 2014). Here, we will have also a loss through evapo-transpiration, potentially obtaining **>95% rainwater reuse** or **minor** in dry climates locations with low rainfall patterns. The service model will consider a payment fee per m^3 water treated and reused. This solution will be developed and demonstrated **in WP4** at lab and semi-pilot scale (TRL 5) previous implementation at large scale (Demo buildings) (TRL7). Operating parameters obtained in lab scale tests with NBS will allow the optimisation of O_3/UV technologies as well as the proper dimensioning of NBS for final integration. NBS will be developed by ALCN. Disinfection using O_3/UV will be optimised at lab and pilot scale by LEITAT and finally integrated in buildings by IDP (See Demo 1 and 2 in Figure 3).

Frontrunner building: Demo 1 - (60's) Demo 2 - (70's) Demo 3 - (1900) Demo 4 -(2017)

Solution 6 (based on **N6**):	**Efficient treatment and reuse of un-segregated water**	Potential type of service:	**Renting/ leasing**
Interested Stakeholders:	St1, St2, St3, St4	Linked building life cycle:	Design, refurbishment and use

WW can be combined in one single pipe with kitchen-shred organic waste including food-waste from all floors of an existing building. The liquid fraction of this combined WW can be separated from the solid fraction with a separation unit using centrifugal force and gravity. The liquid fraction can be treated with NBS indoor, at facade and outdoor with a minimum energy cost (<1.5 kWh/m^3) and allowing **>95% water reuse**. The effluent water can further be disinfected (by using O_3/UV) and used for toilet flushing, urban gardening and compost production. The solid fraction can be used for biogas production in a dry anaerobic digestion unit (Solution 8). The feasibility of this water treatment solution will be demonstrated at laboratory and pilot scale (See Demo 3 in Figure 3). The potential reuse of treated water for toilets or irrigation will be also quantified and demonstrated by ALCN **in WP4**.

Frontrunner building: Demo 1 - (60's) Demo 2 - (70's) Demo 3 - (1900) Demo 4 -(2017)

Solution 7 (based on **N7**):	**Blackwater and BW treatment for biogas production**	Potential type of service:	**Renting/ leasing**
Interested Stakeholders:	St1, St2, St3, St4	Linked building life cycle:	Design, refurbishment and use

This solution includes two core AD technologies for treatment of blackwater and BW at building scale, including the efficient BW collection and grinding at source. Main difference between anaerobic technologies resides in the water content of the effluent to treat. **Solution S7a:** is based on the joint treatment of blackwater and grinded BW at building (pilot) scale for wet anaerobic digestion using AnMBR solution. The solution foresees the recovery **>95% food waste** and **>95% organic matter from blackwater** for the potential generation of (1) **>90% treated blackwater reuse** for toilet or fertigation; (2) **renewable energy from biogas** (linked to S7c), (3) and **compost** (linked to S8) at home level for use by dwellers, neighbours or use in public gardens. In HOUSEFUL, LEITAT will design and construct an AnMBR system adapted for blackwater and grinded BW mixture at laboratory (5L bioreactor) and pilot scale (organic loading rate of 1520 inhabitant eq.) as a compact and transportable unit. The design will be addressed for a minor odour production. The systems will be started up and optimised in LEITAT facilities using real WW and grinded BW effluents. Biogas will be treated using absorption technologies developed by CARTIF (**Solution S7c**) ensuring a biogas quality corresponding to natural gas grid specification for potential grid injection, use in CHP, combustion in a natural gas boiler or other valorisation routes proposed by stakeholders. Once safety of the system is ensured and O+M protocols are prepared, the system will be emptied and transported in a container for final integration in the building (Demo 1) in operational environment at pre-commercial scale. The biogas conversion system will be initially dimensioned for the renewable energy recovery of **2600 kWh/y from biogas**. **Solution S7b**: After separation of the un-segregated WW in Demo 3, the solid fraction and grinded BW can be mixed and used for production of

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biogas and compost using a dry anaerobic digester. Goals foreseen by these solutions include the minimum recovery **>95% food waste** and **>95% organic matter from solid fraction of WW** for production of r**enewable energy from biogas** (linked to S8) and **compost** (linked to S7c) at home level. HOMEBIOGAS system will be adapted for biogas and compost production from BW or manure to the new feed. To make the existing system to work autonomously, some modifications like a feed slide need to take place. The sludge at the bottom of the anaerobic digester can be emptied regularly (once every 6 months), to produce high-quality compost. The generated biogas itself can be cleaned by using the absorption solution developed by CARTIF (**Solution S7c**) for increase of CH_4 content (< 80%) and stored in inflatable and impermeable fabrics. This energy storage can be further used for potential grid injection, use in µCHP, or combustion in a natural gas boiler and can support the NZEB or even passive house claim to reach full autonomy (100%) even in winter times (December-February), when heating systems are mostly needed and sun heating will not be easily available. Demonstration of proposed solutions will be performed **in WP4** activities.

Frontrunner building: Demo 1 - (60's) Demo 2 - (70's) Demo 3 - (1900) Demo 4 -(2017)

Solution 8 (based on **N8**) **High quality fertiliser/compost of local origin** Potential type of service: **Renting/ leasing Capacity building service** Interested Stakeholders: St4 Linked building life cycle: Use

Stabilised BW from anaerobic digestion systems (AnMBR or dAD, Solution **S7**), will be used as compost for local gardening (public parks, training activities in schools, etc.) substituting commercial compost. For that, on the one hand, reliability of different low-cost options for stabilisation of AnMBR digestate (from S7a) such as drying (IDP, LEITAT), domestic co-composting by mixture with fresh BWs (LEITAT) and vermicomposting (LEITAT) will be compared at laboratory (TRL5) before selection optimisation (TRL6) and scale-up and demonstration at pilot scale in Frontrunner buildings (at TRL7 in Demo 1), **in WP4**. See Figure 3. Target foreseen includes the mass conversion yield of **0.55kg compost/kg digestate**. On the other hand, an innovative compost cultivator prototype (developed by ALCN) will cocompost digestate from dAD in S7b solution together with fresh biomass clippings from the garden and greenhouse and will generate plants at the same time in a self-irrigating system. Any odours will also be eliminated by the plants and their roots in the closed system. A liquid fertiliser will be collected at the bottom of the system, hygienised, and can be sold or traded for organic waste with the neighbours. The best service model for this process will be co-elaborated with local and regional stakeholders. This model can possibly be expanded to bigger scale for decentralised urban composting activities. The operational performance of the prototype will be firstly tested at TRL6 before demonstration and installation in Demo 3 at building scale (at TRL7), both activities **in WP4**. Replication of the optimal solution demonstrated in both demos will be offered to stakeholder as renting/leasing service.

Frontrunner building: Demo 1 - (60's) Demo 2 - (70's) Demo 3 - (1900) Demo 4 -(2017)

Solution 9 (based on **N9**): **Optimal management of waste at service the end of building life cycle** Potential type of service: **Customer** Linked building life cycle: Design, refurbishment, dismantling Interested Stakeholders: St1, St2, St3, St4

The Solution 9 is focused on the analysis of maximum recovery and valorisation potential of existing materials in buildings (i.e. envelope, facade, etc) at refurbishment or demolition stage, providing demolition guides to guarantee the best and safety management of waste streams. In HOUSEFUL, this innovative solution will be offered to stakeholders as a service, focused on Design for Disassembly (DfD) principles impacting in the new design of residential buildings with increased longevity by easier renovation potential, using simple and modular building structures and systems during construction phase to make easier and much more efficient the dismantling process. The solution includes the selection of safety procedures for disassembly and reuse of non-hazardous materials in construction elements, aiming to achieve a potential **reduction in 30**% of CDW currently produced during building dismantling/demolition. The DfD will ensure materials are easily reused through the use of non-hazardous and durable materials. Thus, DfD service allows an easier maintenance, repair, recovery and reuse of components and materials that still have economic value for reuse in other buildings. Solution 9 will be demonstrated by ITeC **in WP4** and in all Demos.

Frontrunner building: Demo 1 - (60's) Demo 2 - (70's) Demo 3 - (1900) Demo 4 -(2017)

Solution 10 (based on **N10**)	**Improvement of energy efficiency by active and passive solutions**	Potential type of service:	**Customer service Renting/leasing**
Interested Stakeholders:	St1, St2, St3, St4	Linked building life cycle:	Design, use and refurbishment
Design of energy efficient facades and roofs using modular and more sustainable materials selected according to circular criteria. Assessing on the optimisation of global refurbishment actions according to energy efficient criteria. Including possible energy generation in site of the building, in order to compensate, as maximum as possible, the non-			

renewable primary energy consumption. An energy saving service will be offered to reduce the non-renewable energy consumption in the usage phase of buildings and reducing billing costs to the lowest level. Energy improvement solutions such as envelope interventions (e.g. innovative facades, energy-efficient windows, etc.), solar thermal systems, shared photovoltaic systems, etc will be proposed to reduce the energy demand of buildings and increase the share of renewable energies, contributing to reach Nearly Zero Energy Buildings (nZEB).

The service includes an initial evaluation of the feasibility of the energy solutions proposed as well as energy tips for dwellers to reduce the energy bill, and the best renewable systems to be implemented, according to the building characteristics, the end-users' profiles and the national regulations and energy markets backgrounds. This solution will be demonstrated **in WP4** by AIGUASOL and ITeC, with the collaboration of project partners such as ALCN, IDP, GWR, AHC, NEUS LEBEN and CARTIF.

Frontrunner building: Demo 1 - (60's) Demo 2 - (70's) Demo 3 - (1900) Demo 4 -(2017)

	Guarantee the energy saving/ production in buildings	Potential type of service: Linked building life cycle:	**Renting/ leasing** **Capacity building service** Design, use and refurbishment
Solution 11 (based on **N11**)	☒	☒ ☒	☐
Interested Stakeholders:	St1, St2, St3, St4		

Co-creation based on performance rather than ownership for energy systems (i.e. ST, PV) finally installed in each building (Pay per performance business model). Monitoring of energy saving by active and passive measures adopted from material and energy valorisation points of view. Pay per performance business models will be promoted to be applied in the residential sector, in which the output generated is billed instead of the product ownership. For instance, a manufacturer or a technology supplier leases a technology/equipment and retains their ownership, encouraging manufacturers to design long lasting and highly efficient products and encouraging technology suppliers to provide efficient products. As an example, a pay per performance service applied to renewable energies can consist in that an energy service company supplies, installs and maintains a solar thermal (ST) or photovoltaic (PV) system, guaranteeing an annual solar production defined in a contract of 10 years. By using this model, it is guaranteed that ST or PV systems operate correctly at least during the contract signed between the energy service company and the owner/neighbourhood community of the building. At the same time, these renting and leasing proposed models require from a continuous monitoring not only of the renewable energy production, but also of the energy consumptions of the end-users. By giving them the related information will help to reach the energy savings goals and, by that, guarantee the service in the agreed conditions. This solution will be demonstrated **in WP4** by AIGUASOL and ITeC, with the collaboration of project partners such as ALCN, IDP, GWR, AHC, NEUS LEBEN and CARTIF.

	☒	☐	☒	☒

Frontrunner building: Demo 1 - (60's) Demo 2 - (70's) Demo 3 - (1900) Demo 4 -(2017) Demonstration of HOUSEFUL solutions for water (S5, S6) and waste (S7, S8) vectors requires the technological implementation of solutions in Frontrunner buildings 1, 2 and 3. Figure 3 shows main water/BW flows and planned technologies behind HOUSEFUL solutions.

Improved access to finance through investment planning for enhanced dimensioning of technical specifications and replication of proposed solutions

Further than the EC contribution, other financial resources to create a feasible investment plan for proper dimensioning of technical specifications as well as for allowing the replication of 11 HOUSEFUL solutions marketable as services will be sought. Several options will be analysed such as other national or regional funding schemes, RIS3, crowd funding through ethical banks, bank loans and/or guarantees through blending instruments from EIB (e.g. InnovFin, ESIF) and national banks, ERDF (if applicable). It is also foreseen to explore other alternative financial schemes, mainly those that will use as foundations an innovative public and private business models or even an only private model. New economic resources promoting the final definition and exploitation of HOUSEFUL solutions as services will be welcome and offered for successful replication activities in Follower buildings or new case studies after the end of project execution.

Environmental, economic and social assessment of the solutions demonstrated will be carried out by applying a life-cycle thinking approach.

The achievement of technical objectives foreseen for each HOUSEFUL solution will be supervised by a **Technical Monitoring program,** led by ALCN and developed **in WP4**. The program will include those sensors and protocols to evaluate the correct performance of technological solutions during their demonstration phase in each Frontrunner building. The implementation of solutions in HOUSEFUL will be accompanied by a pro-active market introduction **(in WP6)**, using continuous interaction with stakeholders as a key management strategy along the whole value chain

(in WP3). In addition to its practical demonstrations in Frontrunner buildings, the HOUSEFUL solutions will be environmental and economically evaluated **in WP5** through **LCA and LCC, as key tools to assess the impact of the transition from a linear approach to a circular model.** The overall of proposed solutions in the **4 Frontrunner buildings** will be also simulated with BIM, which will provide key data for conducting the LCA and LCC. In addition, the Level(s) tool developed by the European Commission will be used to assess the environmental performance of Frontrunner buildings with the aim to provide feedback and recommendations for improving the tool. In parallel, the **evaluation of the social impact** will be carried out **in WP5**. This measure will help boosting social understanding through the analysis of stakeholders' attitudes and behaviour and the development and implementation of an engagement strategy aimed at undertaking social interactions. Therefore, HOUSEFUL will assess the 3 pillars of sustainability: environmental, economic and social.

The co-creation of CEBOs will enable to **offer the solutions identified as new systemic services** for the housing value chain. The identification of **market constrains**, social barriers and enables, a detailed definition of the **exploitation plan** and an effective strategy for the engagement of different stakeholders will be key factors to provide circular services. **A Solution-to-Service Monitoring Plan (SSM Plan)** will be developed by LEITAT to monitor the transition process from solutions to services during the project execution.

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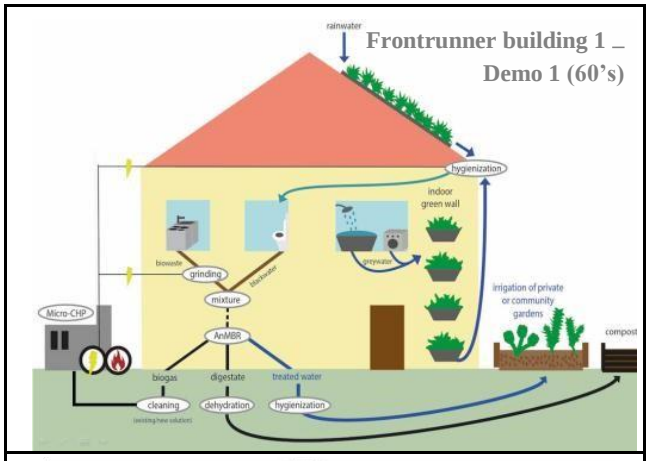
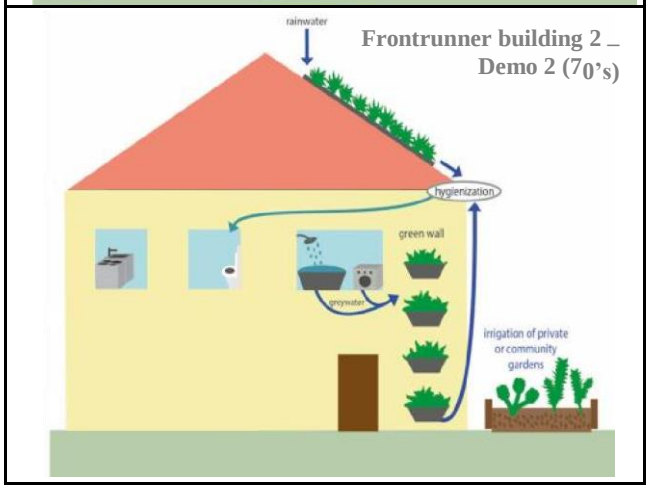
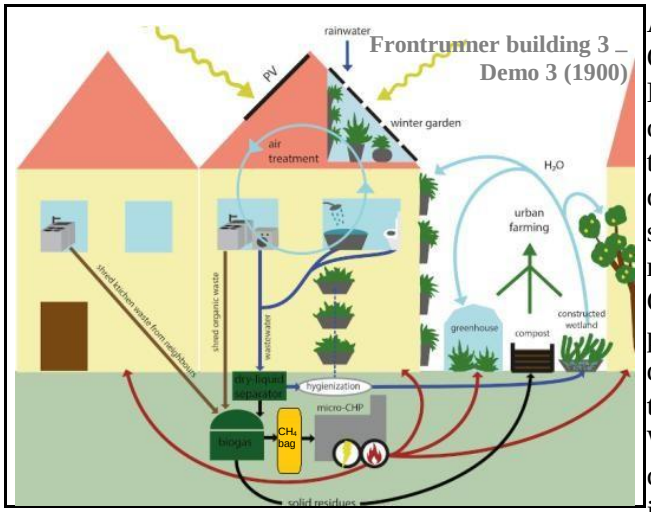
An innovative value added of HOUSEFUL will be the creation of a **Collaborative Community of Housing Experts (CCHE) board.** The CCHE will **accelerate the transition towards a circular economy** in the built environment through the **engagement of 10 additional demo sites (Follower buildings) spread around the world** for the **replication of the HOUSEFUL services** and the dissemination of the results among its network of contacts.







All the previous activities, along with a focused and accurate Communication and Dissemination, will allow a successful Market Uptake. Apart from the previously described complementary competences, project success will be achieved thanks to the suitable selection of HOUSEFUL partners and the collaboration of the CCHE board, which will act as an external scientific-technical panel for the definition of the optimal use of resources during different stages of building life cycle. The CCHE will create synergies with previous and ongoing related projects. Since a multidisciplinary approach is required to define a fully circular model, the **CCHE board brings together experts from different disciplines** such as energy, WW and solid waste treatment experts, architects, engineers, construction companies, technological providers, research institutions and public administration. The board will also include representatives from the HOUSEFUL partners (the Core Group, which will retain the responsibility for scientific coordination, project objectives and the replication of solutions in the follower buildings). The CCHE board will be initially composed by local, regional and national stakeholders from each HOUSEFUL country**,** but open to relevant stakeholders from other EU countries to join discussion panels and roundtables. CCHE board will be involved in ***(1)*** the development and quantification of KPIs defining the new circularity value of buildings, ***(2)*** the definition of the optimal use of resources during different stages of building life cycle, ***(3)*** be the main recipients of project results, ***(4)*** the first actors receiving the status of project performance and ***(5)*** the main driver for replication of HOUSEFUL services and proposed solutions in existing and/or new buildings around Europe and engagement of **10 Follower buildings** for the **replication of proposed services** in Europe. All the potentially interested target groups representing the potential adopters and replicators of the HOUSEFUL services will be approached through dedicated dissemination (**in WP7**) addressed to different type of audience including stakeholders in Frontrunner buildings. The presence of EU and national housing associations in the consortium (CECODHAS, AHC, NEUES LEBEN) will maximise HOUSEFUL results' dissemination potential through their existing communication channels. Project results will be disseminated at a minimum of 30 conferences and fairs, impacting on more than 200 companies.

Figure 3 Main water/waste/energy flows HOUSEFUL solutions: water (S5, S6) and waste (S7, S8)

While there are several instruments which impact on buildings and construction products, such as the Energy Performance of Buildings Directive, the Energy Efficiency Directive, the Construction Products Regulation, the EU Emissions Trading System, the Industrial Emissions Directive, the Waste Framework and the Landfill Directives, they focus on different resources and parts of the life-cycle and are not yet able to provide an overall life-cycle approach (EC, 2014). HOUSEFUL will provide **policy recommendations** (**in WP6**) at local, regional, national and EU scale to strengthen the legal framework with the aim to facilitate the transition in the housing sector towards a circular economy. Existing policies for promoting renewable energies and energy efficiency in buildings need to be further strengthened and complemented with policies for resource efficiency, which look at a wider range of environmental impacts across the life-cycle of buildings. Recommendations for updating local, regional, national and EU regulatory framework will be provided based on project results and lessons learnt. Policy makers will be contacted during the execution of the project for the easier achievement of these actions.

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Policy recommendations to foster circular principles in the housing sector

1.3.1.2 Positioning of the project

The feasibility of HOUSEFUL strategy will be demonstrated and validated in 4 selected buildings (Frontrunner buildings) aimed at increasing their circularity in water, waste, material and energy vectors. While an overall TRL around 7 to 8 will be achieved for HOUSEFUL concept, 8 specific technological innovations will make the jump from TRL 4-5 to TRL 7-8, offering/marketing proposed solutions as new services (TRL8) to stakeholders in housing sector at the end of project execution. In relation to HOUSEFUL innovations, the forecasted TRL evolution of mentioned individual technologies during the project compared to the current one is explained in Table 2.

Table 2: TRL evolution of technological solutions implemented in Frontrunner buildings in HOUSEFUL

Process/ Product	TRL		Description of the technological innovations demonstrated in HOUSEFUL	Beneficiary
	Start	End		

Green façade for water treatment (Solution 5)	5	7	The green façade will be constructed at semi-pilot and pilot scale consisting of more than 70% CDW and is meant for circular building anyways with its modularity and upcycling character. The outer layer will include plants inserted to be used for rainwater and greywater treatment. This facade offers the services of micro-climatic benefits, water treatment and plant/food production.	ALCN
Constructed vertical wetland indoor (Solution 6)	6	8	The vertical ecosystem® (or short vertECO®) cleans greywater in a multi-layer green wall system, which has been developed and demonstrated as prototype in a real environment meant for indoor use with the water flow of up to 2 m^3/day. Being able to offer this solution only for greywater is a big constrain for most households or touristic facilities (which are one target group of this product), due to the need of a second plumbing system, which in most cases do not exist. So, HOUSEFUL plans to separate the liquid from the solid fraction of conventional WW and treat the liquid fraction in subsurface flow. We will also try to combine the lower levels of the treatment area with the production of edible plants. This green wall is not only aesthetic and can support air purification, but it also offers the service of water treatment and food production at home level (TRL8).	ALCN
Interior winter garden for joint treat-ment of air and rain-water/treated WW (Solution 6)	5	7	Currently ALCN develops an air treatment (temperature control, cleaning, humidification) system with plants in a winter garden (RECO_2ST, H2020-EeB-2017). This system will have flaps at the bottom and the top facing inside and one flap towards the outside. Depending on the temperature and the humidity in- and outside these flaps can open and close. The air that passes through the system, is also cleaned. In HOUSEFEUL we will install these prototypes for every level and will foresee automatic feed with rainwwater or treated WW to be completely autonomous. Air treatment will include temperature control, cleaning and humidification through automatically controlled systems whereby plants are fertigated by on site treated waste water. Operational feasibility of installed winter garden will be demonstrated from TRL 5 to TRL 7.	ALCN
Modular and transportable AnMBR for joint treatment of blackwater and grinded BW at building scale (Solution 7)	4	7	AnMBR is a commercial technology used for the treatment of agro-industry streams (TRL9). Companies such as ACWA, ADI Systems, KUBOTA or Veolia commercialise AnMBR systems. But this technology has never been tested and demonstrated for the treatment of WW and BW at building scale. In HOUSEFUL, optimal AnMBR configuration and membranes will be selected for efficient treatment of blackwater (↑COD) and grinded BW (↑COD,↑SS) mixed at optimal ratio at lab scale (TRL4). Operational results using real WW and grinded BW (TRL5) will be the basis for the design and development of a modular and transportable AnMBR at pilot scale for treatment of 8-10 hab-eq blackwater for first operation in LEITAT facilities (TRL6) and further integration demonstration at larger scale (TRL7) in Demo 1.	LEITAT
Dry anaerobic digestion (dAD) for valorisation of BW and solid fraction of WW at resi-dential level (Solution 7)	5	7	HOMEBIOGAS commercialises compact, transportable and modular dry AD aimed at efficiently valorise BW and manure for biogas production (cooking) and compost at building scale (TRL9). HOMEBIOGAS will adapt their design at three family building scale for treatment of the solid fraction of WW and BW. The system needs to work in a semicontinuous mode, being able to be automatically fed with almost dry waste. For the sludge and liquid discharge together we will need to develop a suction unit that works in batch mode and directly feeds the compost unit. The technology will be designed according to energy needs of the Demo 3 but also considering energy/compost needs of the neighbourhood which can provide BW for renewable energy production and receive benefits from their engagement (i.e. compost). The system will be offered as a renting (TRL7) promoting the circular use and transition from ownership to functionality (pay per use).	HOMEBIOGAS & ALCN
Cleaning biogas for cost-efficient CHP conver-sion (Solution 7)	4	7	Gas filters and scrubbers are technologies commonly used in biogas cleaning (TRL9). Many companies commercialise them (Siga-Tech, CJC, Parker, etc.). In the case of membrane contactors, it is a novel technology originally used for liquid gasification although it has been successfully demonstrated in the last years also in biogas upgrading (TRL9). Companies such as 3M or Liqui-Cel commercialise these systems. In both cases, the mentioned technologies have never been tested and demonstrated for biogas valorisation at building scale. In HOUSEFUL, optimal configuration will be selected for efficient treatment of biogas according to the predefined potential final use (CHP, grid injection, transport, etc.) in the different demos.	CARTIF

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Valorisation of biogas by CHP (Solution 7)	5	7	Small biogas CHP units are so far only prototyped and not yet integrated nor verified. An exhaustive cleaning of biogas impurities (CO_2, H_2S, dioxins, etc) could allow its valorisation as natural gas-like fuel. Cleaned biogas can be stored and/or valorised for the production of renewable energy. HOUSEFUL will define the optimal energy valorisation options to cover energy needs of dwellers. For both anaerobic systems demonstrated in HOUSEFUL, the following alternatives will be evaluated: *(1)* in Spain: cleaned biogas combustion in a boiler, valorisation using CHP and power storing in an electric battery, and others; and *(2)* in Austria: Biogas storing in inflatable and impermeable fabrics for further used for potential grid injection, use in µCHP, or combustion in a natural gas boiler Solutions will be tested at TRL 7 and offered as services (i.e. equipment renting) for replication.	ALCN/ IDP
Compost production at local level (Solution 8)	5	7	Several solutions for the stabilisation of BW from anaerobic digestion systems will be tested at semi-pilot scale (laboratory environment) before demonstration and optimisation at building scale (TRL 7) once connected to foreseen AnMBR or dAD technologies in Demo 1 and 3. In detail, final stabilisation techniques for AnMBR digestate such as drying (LEITAT), domestic co-composting by mixture with fresh BWs (LEITAT) and vermicomposting (LEITAT) will be previously studied at laboratory (TRL5) before optimisation (TRL6) and scale-up and demonstration at large scale by IDP (TRL7 in Demo 1). On the other hand, an innovative compost cultivator prototype will be developed at TRL6 and optimised for composting fresh gardening wastes before mixing with dAD digestate at large scale in Demo 3.	ALCN

1.3.1.3 National or International research and innovation activities linked with HOUSEFUL

Most HOUSEFUL partners have coordinated or collaborated in relevant projects with activities and actions linked to this demonstration project, exploring various innovations and circular solutions on water, waste, materials and energy vectors (Table 3). Other related projects and patents are listed in Table 5 and Table 6 to show HOUSEFUL progress beyond SOA.

Table 3: Research and innovation activities providing outputs to the project

Project	Description	Partner	Relevant results
Project demEAUmed	Closing the water cycle in Mediterranean tourist facilities (FP7 Water Inno&Demo, GA. 619116) - Greywater treatment with vertical ecosystem (indoor)	ALCN, LEITAT	Plant consortium for greywater treatment indoor, spatial arrangement in the building
Project Green INSTRUCT	Green INtegrated STRUCTural elements for retrofitting & new construction of buildings (H2020-EeB-04-2016, GA 723825) - Grey- & rainwater treatment by green facade	ALCN, LEITAT	Grey- and rainwater treatment with green façade. Successful anchoring in the facade
Project ReCO2ST	Residential Retrofit assessment platform and demonstrations for near zero energy and CO_2 emissions with optimum cost, health, comfort and environmental quality	ALCN	Indoor air treatment and conditioning with plants and automation in winter garden.
Project Wohnwagon	Includes NBS in the roof of the mobile home for water treatment/reuse. Includes full nutrient cycles, sustainable materials (PVC free, insulation, etc.), farming possibilities.	ALCN	Closing resources-, nutrient- and energy cycles for mobile homes.
4RinEU	4RinEU (H2020) is about Robust and Reliable technology concepts and business models for triggering deep Renovation of Residential buildings in EU.	AIGUASOL, AHC	Buildings energy behavior and BM analysis. Adaptation of the prefabricated facades to demos, including energy impact analysis and BM definition.
InterGRIDy	InteGRIDy (H2020) integrated Smart GRID Cross-Functional Solutions for Optimised Synergetic Energy Distribution, Utilisation Storage Technologies.	AIGUASOL	Link among the detailed solutions and the DHC existing systems.
GrowSmarter	GrowSmarter (H2020) is one of the Smart city projects that aim to transforming cities for a smart, sustainable Europe.	AIGUASOL	KPIs on energetic & economic behavior of buildings used for LCA/LCC.
Plug&Harvest	Plug-n-play passive and active multi-modal energy Harvesting systems, circular economy by design, with high replicability for Self-sufficient Districts & Near-Zero Buildings	AIGUASOL	First approach to end user's requirements, SoA and market, limitations, KPIs and outputs of the circular economy BM to be used in HOUSEFUL
URBAN GreenUP	Aims at obtaining a tailored methodology to support the co-development of Renaturing Urban Plans focused on climate change mitigation and adaptation and efficient	ICONS, LEITAT CARTIF	KPIs, monitoring and demonstration of the effectiveness of NBS; methodology and tools to support deci-
Project	Description	Partner	Relevant resul
	water management through NBS.		sion-makers

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STARDUST SCC-01 Lighthouse project (2017)	Transformation of carbon supplied cities into smart, high efficient, intelligent and citizen-oriented cities, developing urban green solutions and business models, integrating buildings, mobility and efficient energy through ICT.	ICONS	KPIs, monitoring and social impacts through co-creation and engagement of citizens and stakeholders; replication and business models
Project H2020 AfriAlliance	The Africa-EU Innovation Alliance for Water and Climate, building an alliance of stakeholders across Africa and Europe in Water and Climate issues and setting up communities of practice on specific local problems	WE&B	Stakeholders mapping, analysis and visualisation based on social network theory.
GGGI Peru	Pilot testing the Green Growth Potential Assessment (GGPA) in Perú to support the government in carrying out a gap analysis as a preliminary step toward the formulation of the National green growth strategy	WE&B	WE&B analysed the results and contributed to the final report of key recommendations for future Green growth interventions.

1.3.2 Methodology

1.3.2.1 Overall methodology

HOUSEFUL activities have been structured as interrelated Work Packages described as follows:

WP1. Project management

LEITAT will be responsible for the **coordination and monitoring** of the overall operation and progress of the project with the support of the Technical Committee to ensure that all planned activities are properly executed. All partners will collaborate in order to: i) ensure that the milestones and WP deliverables are met on time and quality; ii) assist to project meetings, and iii) favour a fluent communication between partners and WPs.

WP2. Methodology for the evaluation of circularity in the housing sector

WP2 involves the **development of** a robust and reliable **methodology to quantify the circularity level of residential buildings** based on calculation procedures and KPIs (environmental, economic and social), taking into account the whole life cycle of buildings. The methodology developed in this WP will be the basis to create the Circularity Tool (CT), which will be integrated in the SaaS (WP6). Existing market circular solutions and services will be researched in WP2 with the aim to be integrated as part of the SaaS.

WP3. Effective **social engagement: starting the cocreation of HOUSEFUL services**

Based in a stakeholder analysis mapping the structures underlying behavioural and attitudes, WP3 will develop an **engagement strategy** to involve stakeholders from Frontrunner buildings in **the co-creation of CEBOs and solutions** that will be implemented in demo buildings (WP4). Thus, owners, dwellers, tenants, but also design teams (architects, engineers, etc.), municipalities, public housing agencies, manufacturers and technology suppliers will be involved in co-creation activities. Co-creation workshops will be organised to identify community knowledge needs and potential perceived barriers and enablers, as well as to inform about the benefits of implementing circular solutions.

WP4. Large Scale Demo Buildings

WP4 demonstrates the integrated systemic HOUSEFUL service in four Frontrunner buildings, all equipped, adapted and prepared for the **integration of existing and innovative technological and non-technological circular solutions**, covering materials, water, waste and energy aspects. After conducting an exhaustive circular economy audit to determine the initial status of the buildings, the **technical specifications and execution projects** of the solutions to be demonstrated will be produced. Since **innovative technologies** for the treatment of household effluents and BWs will be demonstrated, validation tests will be conducted firstly at lab scale in order to adjust and optimise the proper configuration and operation of proposed systems, and then deployed at full scale in Frontrunner buildings (TRL7-8) once approved permits for technological implementation in Demo buildings are obtained. The performance of the HOUSEFUL solutions will be **monitored** by means of a technical monitoring program defined by project partners for ensuring the quality of the process and the obtained results. Sensors will be installed (photometric sensors, online microbial sensors, humidity sensors, temperature sensors, etc.) when required to monitor the solutions.

WP5. Environmental, economic and social assessment

An environmental, economic and social assessment of the circular solutions demonstrated in Frontrunner buildings (WP4) will be carried out by applying a **life cycle thinking approach**. First of all, an **assessment plan** will be developed to provide a common framework and guidance for evaluating the **environmental, economic and societal impacts**. The environmental and economic assessment will be carried out based on the BIM results (WP4) which

integrate **LCA and LCC** related data and the methodology developed in WP2 for evaluating the circularity level of buildings. In addition, the new Level(s) tool will be tested with real data from HOUSEFUL demo buildings to provide feedback and recommendations to the European Commission for improving the tool. In parallel, **a social assessment** will be carried out for a better knowledge of how the integrated systemic HOUSEFUL service and solutions are going to be received in the socAssociated with document Ref. Ares(2018)1767830 - 03/04/2018iety. The impacts of the baseline scenario will be compared with the new circular HOUSEFUL scenario.

WP6. HOUSEFUL Solutions as Services, market, policies and exploitation

To achieve the expected project impacts, the exploitation activities will be carried throughout the project duration under the leadership of LGI and the support of all partners. WP6 includes a **market entry strategy** of the most promising HOUSEFUL solutions as an integrated systemic service to facilitate the transition from a demonstration phase to a commercialisation phase. **A Solution-to-Service Monitoring Plan (SSM Plan)** will be developed to monitor the transition process from solutions to services during the project execution. A multi-level perspective will be used as a basis for the evaluation of the disruptive **impact of the HOUSEFUL service on existing business models**. Inputs from WP3 (social engagement), WP4 (Large scale demo buildings), WP5 (environmental, economic and social assessment) and from the market study will be used to identify key drivers and barriers to the deployment of the integrated HOUSEFUL service within the current linear business models. The HOUSEFUL service to be exploited will be integrated in a **SaaS**, which will facilitate the shift from linear to circular business models **offering new CEBOs to the stakeholders** of the housing value chain. The SaaS developed will include a Circularity Tool (CT) based on the methodology defined in WP2 and an interactive sharing platform to exchange successful circular solutions. This WP involves the implementation of a **Business Plan** for the HOUSEFUL service for defined market segments and regions as well as the innovation management with relation to the **IPR** and project outputs protection/exploitation. Finally, **policy recommendations** will be provided at local, regional, national and EU scale in order to speed up the transition towards the circular economy in the housing sector.

WP7. Communication and Dissemination

Success stories and lessons learned will be transmitted in WP7, implemented through a **Communication and Dissemination Plan**. Developed knowledge and experience will be communicated to stakeholders and public through different **outreach activities** (workshops, videos, website, webinars, study tours to demo sites, etc), with the contribution of all project partners and the leadership of ICONS. Specific capacity building activities will be organised with the support of the **Collaborative Community of Housing Experts (CCHE) to engage 10 Follower buildings** to replicate the HOUSEFUL solutions demonstrated in WP4 and offered as a service in WP6. Dissemination actions are targeted to achieve social acceptance and end-user involvement, to promote market penetration.

1.3.2.2 Gender analysis in HOUSEFUL Project

Each project partner will implement a Gender Action Plan to promote equality in all possible forms; similar to the ones practised by some of the partners. In LEITAT, AHC, CECODHAS and ICONS more than half of the personnel involved are female. To contribute to the EU Strategy for Gender Equality, the Action Plan will include the following measures: ***1)*** Enforce gender equality in terms of human resources taking into account salary and working conditions; balance between the experts who participate in workshops, training sessions, seminars, etc.; and ***4)*** Balance and monitor gender aspects on a regular basis to tackle the gap for an "equal-opportunities

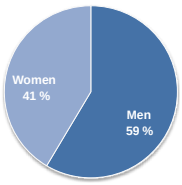

Figure 4. Percentage of women-men staff in HOUSEFUL

[1] ***.4.1 Advances that HOUSEFUL will provide beyond the state-of-the-art***

HOUSEFUL offers an innovative integrated systemic service **for the effective transition from linear model to a circular model** in the housing sector. This service is composed of **11 circular solutions as new services co-created by stakeholders** in current housing value chain. Circular solutions will be demonstrated in 4 European Frontrunner buildings. The integrated service pursues the circular management and efficient use of water, waste, energy and material resources in housing sector. To achieve this new paradigm in the co-creation of CEBOs in the housing sector, main challenges identified in Figure 1 are tackled as follows:

Related to Challenge 1: HOUSEFUL approach fosters a new form of CEBOs' co-creation, increasing the collaboration among stakeholders of the housing value chain. HOUSEFUL provides a new approach in the development of

2) Promote the recruitment of women executive and management positions in project implementation; ***3)*** Maintain a gender

policy". Efforts will be made for better reconciliation between work and family life. With these measures HOUSEFUL will contribute to alleviating gender-based disparities and promote gender equality in accordance with the EU Strategy for equality between women and men (EC, 2010). In order to eliminate gender bias, as a strategic priority, the consortium will take active measures to ensure that women are well represented in the partner's organisations. Opportunities for mobility within the programme will take account of the different needs of women and men in order to enhance participation by female scientists. This approach is coherent with EC policy: The Strategy for equality between women and men (Strategy 2010-2015), adopted by the EC on 21st September 2010, that considers that inequalities between women and men violate basic Human Rights and highlights the contribution of gender equality to economic growth and sustainable development. In HOUSEFUL (Figure 4) project gender equality will exist since already in the proposal phase 41% of the staff involved in the project are women. **The key roles are led by women**: Administrative and Budgetary Manager (Serena Salerno, LEITAT), Technical Manager (Marta Escamilla, LEITAT) and Communication and Dissemination Manager (Elisabeth Schmid, ICONS). In addition, Mrs. Sorcha Edwards (CECODHAS) will be the chair of the CCHE.

1.4 Ambition

proposed solutions, combining technological and non-technological innovative actions, organisational, societal and

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behavioural innovations aiming to promote leasing and/or renting models of solutions as new services, promoting the performance/functionality of products rather than mass customisation.

Related to Challenge 2: Current standards for the evaluation of building circularity are focussed on partial linear processes of the overall building life cycle, on specific vectors (i.e. the cost optimal procedure established in EPBD 2010 for the energy vector) and/or in limited data and quality labelling related to specific materials (i.e. C2C labelling). In HOUSEFUL, a robust, reliable and day-to-day market usable methodology will be developed to quantify the degree of circularity of buildings which could be used by designers but also construction companies, promoters and financial institutions. The methodology will be developed considering the circular economy principles related to recyclability, reusability and waste savings of the materials and active solutions, and also other key factors such as the feasibility of HOUSEFUL circular solutions. The methodology will consider all the existing or brand new methodologies related to the key vectors of circularity, including LCC, LCA, S-LCA methodologies or the next future brand new EU Level(s) one. The methodology will focus on the establishment of replicable calculation procedure, validated on a set of building archetypes representing the main residential typologies for different European geographical locations.

Related to Challenge 3: In HOUSEFUL, potential environmental and socioeconomic impacts of the circular strategies demonstrated will be evaluated by applying a Life Cycle Thinking approach: LCA, LCC and social-LCA methodologies.

Related to Challenge 4: Based on the findings obtained through the analysis of stakeholders' attitudes, beliefs, behaviour and concerns, HOUSEFUL will define an engagement strategy and thus, will organise several capacity building activities (technical workshops, training sessions, study tours to demo buildings) which will help to overcome existing cultural and social barriers for fostering the co-creation of new circular solutions and services.

Related to Challenge 5: HOUSEFUL will review current EU policies and Communications dealing with circular economy, resource efficiency, energy efficiency and renewable energies with the aim to provide policy recommendations able to speed up the transition towards a circular economy in the housing sector. Recommendations to launch circular procurement and circular tendering processes for public social housing will be provided to scale-up new business models. The SaaS developed will be also a powerful tool to facilitate this transition in the housing value chain.

Some HOUSEFUL partners (i.e. LEITAT, CARTIF, AIGUASOL, ALCN, AHC, TURNTOO) have participated in previous EU projects aimed to improve the sustainability of buildings, some of them focused on energy efficiency (4RINEU, Plug-N-Harvest), wastewater (RUN4LIFE) and materials (Urban GreenUP, Green Instruct) in a separated way. This previous partner experience orients the proposal of HOUSEFUL beyond the State of the Art (SOA), with the final aim to offer an integrated systemic circular service for the housing sector. By overcoming the insufficiencies and difficulties of earlier projects, improvements to the technologies and solutions have been defined, as well as organisational and social innovation actions. They will provide **competitive and holistic demonstration of an integrated systemic circular service for the housing sector with optimal, proven and accepted performance.** Table 4 refers to the ground-breaking nature of the objectives, concepts involved, issues and challenges to be addressed, as well as approaches that will be used in HOUSEFUL compared to current state of the art. HOUSEFUL

will give response to the following requirements for a more circular housing sector: i) designing new and more efficient processes and procedures for the construction/refurbishment and demolition phases; ii) efficient treatment of BW and wastewater; iii) reducing the non-renewable primary energy consumption of buildings; iv) making the strategy cost competitive and close to market by including LCA, LCC and social assessment, a new business model and prospective end-users participation, among others.

Table 4: Progress beyond the state of the art and main breakthroughs.

Co-creation of new circular solutions and services	
SOA	• Most of the stakeholders of the housing value chain advance their own interests independently, which prevents the transition towards a CE. These stakeholders need to build new collaborative networks for value co-creation. • Currently, circular practices in the housing sector tend to occur at individual levels (i.e. energy efficiency, materials, waste and water), without considering an integrated approach of the different environmental aspects that cause an impact along life cycle of buildings.
HOUSEFUL approach beyond the SOA	• HOUSEFUL will develop a conceptual framework for shaping the linear systems towards circular economy and will propose how value circles and co-creation of value with a variety of stakeholders are crucial aspects in enabling circular economy. HOUSEFUL will provide a coherent vision for a collaborative network than can translate circular economy principles into housing value chain-specific processes and dynamics. • A social engagement strategy will be carried out for the co-creation of circular solutions and services.
Breakthrough	HOUSEFUL will organise a wide variety of activities (12 co-creation workshops, 4 technical workshops, 2 training sessions, 4 study tours) at EU level to foster the co-creation of new circular solutions and services with the aim of gathering the users' needs, identifying potential barriers for the co-creation of solutions as well as raising awareness about the potential benefits of implementing circular solutions.
A SaaS to bring buildings into the circular economy and promote the replication of HOUSEFUL solutions as services	
SOA	• The full transition from linear to circular models in the housing value chain is still far from being achieved. • Lack of tools in the market able to quantify the circularity level of residential buildings considering different environmental aspects (energy, materials, waste and water) and based on a life-cycle thinking approach, while at the same time offering successful circular solutions and services to the housing value chain.
HOUSEFUL approach beyond the SOA	• A robust and reliable methodology to quantify the circularity level of residential buildings will be developed. The EU Level(s) tool will be considered. • A new circular thinking approach will be provided as a starting point to introduce circular economy and life cycle thinking into building projects, thus raising awareness of stakeholders.
Breakthrough	• A SaaS will be developed and offered to stakeholders of the housing value chain to promote the redesign of traditional business models towards circular ones, encouraging professional development and life cycle thinking. • The SaaS will be a guidance tool composed by a CT to quantify the circularity level of buildings considering water, waste, material and energy vectors. • The SaaS will include an interactive sharing platform to share successful circular HOUSEFUL solutions in residential buildings as new services offered to stakeholders in housing sector, thus promoting the replication of this circular model in other buildings while raising awareness and knowledge sharing activities on circular economy models.
Efficient processes and procedures for the construction/refurbishment and demolition phase	
SOA	• Most of stakeholders in housing value chain does not have detailed information of buildings characteristics to guarantee the potential reuse or substitution of materials while guarantee the functionality of existing elements. • New buildings are being introduced MP and BIM models in design phase, in order to increase the knowledge of their characteristics, enabling the potential sharing of information with stakeholders in housing value chain while potentially increasing their circularity by an improved decision-making. • Lack of information regarding procedures used during construction of existing buildings impacts on the potential reuse of materials during disassembly/demolition of existing buildings.
HOUSEFUL approach beyond the SOA	• MP and BIM offer detailed information of building characteristics, allowing the definition of circular strategies for the improvement of the circular management and use of water, waste, energy and material resources. • In HOUSEFUL, analysis of alternative and safety methods contributing to maximum material recovery during building disassembly for refurbishment or partial renovation while minimising energy consumption and waste production will be carried out.

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Breakthrough	• MP will be developed for selected Frontrunner buildings including a detailed inventory of their physical state of and the elements and materials it consists of by using BIM. A detailed identity of all construction materials used will be achieved. This solution will be offered to stakeholders as a service and will allow the quantification of KPIs for the evaluation of building circularity and detail the technical specifications of HOUSEFUL solutions. • In HOUSEFUL, an innovative service will be demonstrated focused on the Design for Disassembly (DfD) principles impacting in the new design of residential buildings for an easier renovation. DfD includes simple and modular structures and systems to make easier and more efficient the dismantling. DfD will contribute to minimise current CDW disposed to landfill from 40% to 10% in 10 years for housing sector as realistic scenario.
Efficient treatment and valorisation of the organic matter contained in BW and blackwater	
SOA	BW is used to be collected for transportation and further treatment in centralised composting or AD plants. This treatment model generates a high cost for society, making difficult the participation of citizens in the decision making about how to use and exploit the resulted renewable energy and compost for own benefit.
HOUSEFUL approach beyond the SOA	• HOUSEFUL offers decentralised solutions for the successful treatment of segregated BWs and blackwater produced at building level. These solutions give to stakeholders in housing sector the opportunity to generate renewable energy for several uses and compost for gardening purposes. • An innovative AnMBR system will be developed and demonstrated at large scale for the efficient treatment of blackwater and BW produced by 8 homes. Treated WW will be hygienised (O_3/UV) for reuse in toilets. Resulted biogas will be treated with a small scale adsorption system and CHP or heat will be produced for use by dwellers. Social engagement will contribute to decide how to exploit the renewable energy. • Besides, DryAD will be also demonstrated for the treatment of the solid fraction of un-segregated WW streams produced at household level together with BW.
Breakthrough	• AnMBR systems are being operated at large scale in agrifood industries. In HOUSEFUL, a smaller scale AnMBR for treatment of WW streams at building scale (8 homes equivalent) will be demonstrated in Demo 1. • In HOUSEFUL, HOMEBIOGAS will adapt its existing DryAD design at three family building scale for the joint treatment of the solid fraction of WW together with kitchen wastes. The technology will be designed according to energy needs of the Demo 3 but also considering energy/compost needs of the neighbourhood which can provide BW for renewable energy and receive benefits from their engagement (i.e. compost).
Domestic WW treatment and reuse	
SOA	• Although water reuse is an established practise in some European countries like Spain, obtaining reclaimed water from WWTPs (which is the common strategy) results in a big limitation due to the distance from application sites, and the threat of industrial pollutants usually present in centralised systems. • Reclaimed water from WWTP is usually gained from tertiary treatments, once the nutrients (N, P) have been removed. However, as > 70% of water reuse is irrigation in agriculture, these nutrients should remain in water.
HOUSEFUL approach beyond SOA	• In HOUSEFUL, rainwater, greywater and blackwater will be treated by using Natured Based Solutions (NBS) for obtaining reclaimed water for reuse in toilets or irrigation, ensuring its safety by new disinfection by commercial UV/O_3. Micropollutants and pathogens control to evaluate the toxicity to vegetable species in NBS. • In detail, the feasibility of the following NBS technologies for >90% water reuse in toilet recharge or irrigation will be demonstrated at building scale: ***(1)*** Green facades made by CDW for the treatment of greywater and rainwater (Demo 1 and 2); ***(2)*** Vertical ecosystem for the treatment of liquid fraction from un-segregated domestic WW (ver-
	tical wetland indoor) (Demo 3); and ***(3)*** interior winter garden for joint treatment of air and rainwater/treated WW (Demo 3). A final disinfection step will b required for water reuse.
Breakthrough	• Modular NBS and O_3/UV system demonstrated in HOUSEFUL for domestic WW treatment (segregated and unsegregated streams) will be offered to stakeholders as a service (renting/leasing, pay per m^3 treated and reused) • Reclaimed water will be tailored to end-users necessities. For instance, in Demo site 1 and 2 (Sabadell and Terrassa, Spain), reclaimed water will be reused for toilet flushing. On the other hand, in Demo 3 (Baumgarten, Austria), it will be used for toilet flushing and irrigation (in winter garden and external garden).
Energy efficiency and renewable energies	
SOA	• Many EU projects have been conducted so far focusing on reducing the energy demand and increasing the share of renewables in buildings with the aim to achieve NZEB. • New business models are needed to be applied in the residential sector (i.e. pay per performance).

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HOUSEFUL approach beyond the SOA	• HOUSEFUL will design energy efficient facades and roofs considering a circular approach, where modular, prefabricated and off-site construction, design for disassembly, materials reuse and recycling aspects will be considered. • HOUSEFUL includes AnMBR and dAD solutions for blackwater and BW valorisation, promoting the production of renewable energy at building scale. • HOUSEFUL will look into the possibility to boost the use of innovative services based on pay per performance BM, where energy savings or renewable energy production are guaranteed by contract. This type of BM is already applied in industries and non-residential buildings, but implementation in residential buildings is scarce.
Breakthrough	Different energy efficient solutions and services will be demonstrated in Frontrunner buildings aiming to validate their technical, environmental, economic and social feasibility, reducing the energy demand up to 50%.
Development of circular tendering processes for public housing	
SOA	The current setup of tendering procedures launched when constructing new buildings or refurbishing existing ones does not stimulate circular economy principles. Tendering processes are too prescriptive, which results in business not being able to deliver the best solutions and creating little room for innovative solutions.
HOUSEFUL approach beyond the SOA	• HOUSEFUL approach pursues the public sector to lead the way in delivering resource-efficient buildings, as a way to motivate and inspire stakeholders of the housing value chain to apply circular principles in their projects. HOUSEFUL will encourage municipalities and public housing agencies, as parties in charge of initiating building projects, to set the right example when constructing or refurbishing buildings. • HOUSEFUL will increase knowledge on circular economy and how it can be adapted to public procurement.
Breakthrough	The tendering processes that will be launched in 2 of the Frontrunner buildings by AHC will serve as an example of a whole new approach since circular principles will be considered. These experiences and the knowledge obtained will benefit not only the social housing agencies involved in the project (AHC and NEUES LEBEN), but also other public housing agencies willing to embed tendering processes that boost buildings'circularity. At least 8 public housing agencies will be engaged during the project to promote the launch of circular tendering processes.

Table 5: HOUSEFUL progress beyond existing projects.

Project Name, approach and technology	Differences in HOUSEFUL project
BAMB project (H2020) enables a systemic shift in the building sector by creating circular solutions that increase the values of building materials. BAMB develops and integrates tools that enabling the shift: Materials Passports and Reversible Building Design. The BM enables waste reduction and use fewer resources during construction.	BAMB shares with HOUSEFUL the circular economy approach, but only materials are tackled. HOUSEFUL focuses on being more resource efficient in the management of energy, materials, water and waste at residential building scale, including social housing
The 4RinEU project (H2020) will define robust, cost-effective, tailorable deep renovation technology packages supported by usable methodologies, feeding into reliable business models. It will minimise failures in design and implementation, from the preliminary audit up to the end-of-life. It will provide information on energy, comfort, users' impact, and investment performance.	HOUSEFUL will implement a specific software at demo sites that will permit to quantify building circularity based on standard methods. This will result in an innovative methodology that will be shared among interested audience by means of technical workshops and other engagement activities.
During the IEE project TABULA, residential building typologies have been developed for 13 European countries. Each national typology consists of a classification scheme grouping buildings according to their size, age and further parameters and a set of exemplary buildings representing the building types, which enable a projection of the actual national building stock consumption and the energy saving potentials.	HOUSEFUL will consider active and passive efficient solutions with the aim of improving building energy efficiency. To achieve this, specific facades and roof will be designed according to circularity criteria when selecting components and materials in the construction object. During refurbishment stage, best energy efficient actions will be detected. A step forward in reducing energy needs, will be the in situ generation of energy in the building facilities while treating domestic WWs together with organic waste.
The Green INSTRUCT develops a prefabricated modular building block that is superior to conventional precast reinforced concrete panels, improved acoustic and thermal performance and multiple functionalities. The Green INSTRUCT block consists of over 70% of CDW.	
The demEAUmed project demonstrates innovative technologies for optimal and safe closed water cycle in Mediterranean tourist facilities, leading to their eventual market uptake. As well as, the reduction of fresh water consumption by using alternative water sources or the reuse of treated grey waters and/or WWs within the building.	As for demEAUmed project, in HOUSEFUL, grey-water will be treated separately with a tailor-designed NBS technology and a hygienisation technology before water reuse based on O_3/UV. Also, rainwater will be conditioned with these technologies. This will be the first demonstration of the feasibility at residential
The UVOX-Redox combines the oxidising effect of $O_3$with UV dis-infecton and highly effective advanced oxidation process (AOP) in one single system for treating different WWs.	building scale. A 95% water reuse is foreseen, for irrigatio and toilets use.

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Living Fuels is a UK based company that recovers used cooking oil into an environmentally friendly bioliquid which is used to generate carbon neutral electricity for residential homes and businesses.	Blackwater and BW will be jointly treated using a compact tailor-made AnMBR and dAD reactors. The recovered biogas will be cleaned in-situ for (1) storage during winter for further valorisation aor (2) used for direct electricity and heat production, reducing building energy needs. Resulting digestate will be treated for use by local community as fertiliser. Treated blackwater from AnMBR will be disinfected for reuse.
The BIOPAQ®UBOX, technology was designed by Paques for municipal/industrial applications, combining biological sewage treatment with biogas purification. This leads to low operational costs, a small footprint and zero odour emission. The system modular design is suitable for capacities from 5 to 40 thousand population equivalent.	

1.4.2 Innovation potential of HOUSEFUL

Table 6: Results of patent search.

Patents	Approach and technology	Differences to HOUSEFUL
Nr. 516363 Austrian Patent (ALCN)	Greywater treatment with vertical ecosystem "vertECO"	WW treatment including liquid phase of blackwater after hygienisation. No direct blackwater treatment.
US13141082 (Huge Asia Ltd)	Multi-phase separation, generating sewage gas, and biological denitrification /dephosphorisation	HOUSEFUL will produce biogas and treated waste in modular units.
CN 203429173 U Novel domestic biogas digester (孔远萍)	Characteristics of simple and convenient structure, low construction cost, small floor area, strong fermenting capacity, high pressure and safety and reliability in operation	In HOUSEFUL, an optimised and compact dAD and AnMBR for blackwater and BW treatment will be implemented.
CN202643583 (U) (Daqing Yulu Agriculture Technology Dev Co Ltd)	House sewage disposal recovery fertiliser making device. Utilises agricultural waste (in combination with a poultry house)	HOUSEFUL will produce compost from digestate obtained in the AD of blackwater and kitchen waste
CN106594886 (A) (Hangzhou Jiegao Env Prot Tech Co Ltd)	System and method integrating air purification with plant cultivation	HOUSEFUL will also provide WW treatment capacity.
US 20110259832 A1 (Veolia Water Solutions)	Treatment of municipal or industrial WW comprising a step of oxidation via O_3 and another of disinfection via UV radiation	HOUSEFUL envisages using small scale O_3/UV systems-based technologies available in the market to disinfect treated domestic WW prior to reuse at home level for toilet recharge or irrigation.
US6200466B1 (New Lasers Inc)	Water treatment by photolytic reaction	
WO 2010000798 A1 - Waste water treatment (Unilever Ltd.)	Treatment of industrial waste with transition metal catalysts, H_2O_2 and a reactive radical.	

2. IMPACT

2.1 Expected impacts

HOUSEFUL intends to develop and implement circularity principles in the built environment to see buildings as resource banks – demonstrated at large scale in 4 demo buildings and followed up in 10 additional buildings. The market success of the 11 circular solutions co-created between stakeholders is favoured by implementing technological, organisational and social innovation through the whole value chain of the building life-cycle, including the endusers perspective along the project. HOUSEFUL will make possible the transition to a Circular Economy of buildings incorporating four dimensions: economic, social, legal and environmental, through a systemic innovation approach including all the aspects depicted in Figure 5. Concretely, HOUSEFUL will address the expected impacts (EI) in the call as specified below.

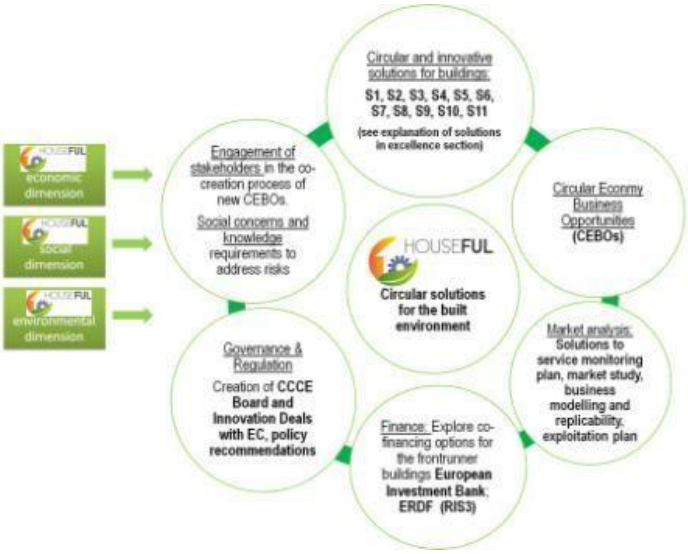


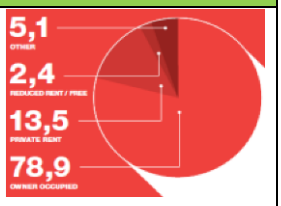
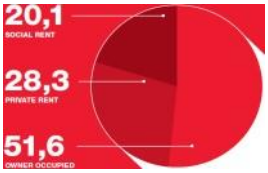
Figure 5. Systemic approach of HOUSEFUL to achieve circularity principles in the build environment based on the EC Action Plan.

EI 1&2: Markets for new products/services (…) more sustainable consumption patterns and innovative producer responsibility or (cross) sectorial governance schemes.

New approaches for designing products/services considering end-users, tenants, brand owners, entrepreneurs, and researchers, and deliver the needs of end-users;

First market niche for the application of HOUSEFUL services is **social housing** (three of the demo buildings are for social housing). The co-created services offered in HOUSEFUL for this sector are presented as CEBOs, which are based on **customer services**, **capacity building services** and **renting/leasing** of the innovative solutions for the whole life cycle of buildings. More precise information of the 11 CEBOS are shown in Table 1 with the respective explanation of the identified needs and vectors tackled. **Current status of EU social housing market** (*based on the report: **The state of Housing in the EU 2015** published by Housing Europe (next report is foreseen in October 2017*[Associated with document Ref. Ares(2018)1767830 - 03/04/2018]). Two facts summarise the current social housing state in EU: a) There are more people without a home today in Europe than six years ago, and 2) there are not enough affordable homes available in most European countries to meet increasing demand. Additionally, in many countries the quality of the buildings is rather poor (especially in central and Eastern European countries). For example, **running water is not accessible in 1 out of 3 homes** in Romania and only **65% of households have a WW system**. With this scenario local governments and their respective social housing agencies are currently setting up programmes for urgent renovation of existing buildings (HOUSEFUL demos 1&2 in Spain) or even investing in new construction of social houses (HOUSEFUL demo 4) Table 7 summarises the Spanish and Austrian housing market status. With this scenario together with the identified needs in Table 1 is clear that the social housing sector demands for the creation of new services to foster sustainable consumption patterns during the construction, renovation and using phases.

Table 7. Trends in social housing markets in the countries of HOUSEFUL frontrunner demo-buildings

Trends in housing market – Spain (data 2015)[ii]	
Total nº of dwellings (thousand): 25,208 Nº of dwellings/1000 inhabitants: 538 **Advantages: (1)** Social housing funds: banks transfer a proportion of their housing stock into the social rental sector; **(2)** State Housing Plan subsidizes creation of public social rental housing. **Disadvantages: (1)** Numerous empty homes owned by banks; **(2)** Extremely high number of foreclo- sures undertaken from 2007-2012	
Trends in housing market – Austria (data 2015)	
Total nº of dwellings (thousand): 4,441 / Nº of dwellings/1000 inhabitants: 555 Total housing completions in 2012/13: 52,000; Total nº of social rental dwellings: 891,000 **Advantages: (1)** High quality of dwellings, large living space; **(2)** Stable housing market little impacted by the global financial crisis **Disadvantages: (1)** In terms of rent affordability, share of housing costs out of disposable income and housing overburden rate, Austria scores better in terms of share of housing costs than EU average, but slightly worse if we look at low income population	

i) Identification and analysis of stakeholders' belief structures and its involvement through HOUSEFUL project and its co-creation process

The HOUSEFUL Service will benefit the following stakeholders (St), which will be the essential elements in the cocreation process of HOUSEFUL solutions towards the adoption of more sustainable and responsible choices for their business and decision-making processes. The solutions will also address factors conditioning social choices and influencing behavioural intentions of stakeholders. The following groups of St (described in p. 2) will form the CCHE: **(St1) Construction companies, private promoters, technological and product suppliers** will be engage to redesign their linear products/services production towards circularity principles. **(St2) Design teams including architects, engineers and quantity surveyors** will participate in HOUSEFUL co-creation workshops and study tours to guide towards the circularity potential all buildings and encourage them to set up housing plans and projects considering such potential. **(St3) Municipalities and regional/national housing agencies** will be engaged towards the adoption of circular criteria in their tenders and housing local programmes of renovation and new construction of buildings. **(St4) Owners/Dwellers/tenants**: this group of stakeholders has key importance in the successful implementation of HOUSEFUL services, thus their involvement, and engagement from the very beginning is essential for the project's success. (**St5**) **Policy makers and implementers agencies**: CCHE Board will count with policy makers around the housing sector to provide them with sustainable guidelines and policy recommendations on how to design a strategy to accelerate the transition towards the circular economy.

ii) Implementation of innovative producer responsibility and cross-sectorial governance schemes

Although sustainable building environment initiatives have already been partially established, the EC has identified challenges such as the changes in energy efficiency, more active uptake of new technologies and services, improving specialised training, and stimulating demand for efficiency improvements in existing building and renovations. This will require innovative manufacture responsibility and governance schemes across the stakeholders identified above to enable the housing sector to become more competitive, resource efficient, socially inclusive and sustainable.

HOUSEFUL will enable the implementation of the proposed innovative services in 4 buildings involving not only the producers of materials/products and technologies but also directly the civil society, decision makers and public housing agencies.

EI 3 & EI 6: Reduce supply chain length, increase resource efficiency, and reduce environmental impact. **EI 6**. Economic, social, cultural and environmental sustainability and main elements of **business plan**. Assessment of possible positive and negative side-effects and risk and harmful substances present in recycled materials.

As previously mentioned, the housing sector is one of the highest consumers of resources in EU, accounting for **50% of extracted materials**, **42% of energy consumption, 35% of GHG emissions**, **30% of water consumption** and **35% of total waste production**[3]. HOUSEFUL approaches the environmental, social and economic aspects in a holistic manner: Sustainability assessments will be based on a LCA, LCC and Social-LCA that will enable detection of the positive and negative impacts of the HOUSEFUL services. Moreover, the project guarantees to reduce and close the current supply

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chain length of the WW and BW services as well all the services along the construction phase of buildings. This will be possible by the creation of commercial software as a service to calculate the Circularity Level of the demo buildings and of at least 10 additional follower buildings. This SaaS will contribute to the utilization of "**Level(s)**", the new tool developed by the European Commission together with industry stakeholders in order to enables building professionals and their clients to identify key areas for better environmental performance. The inscription to "Level(s)" register will be proposed and warmly advice to housing stakeholders during the dissemination activities. Reduction of the housing vectors supply chains will contribute to these main expected benefits:

ECONOMIC BENEFITS (also addressed in EI 1 & EI 5)

market opportunities	HOUSEFUL provides new opportunities for (social) housing sector by offering new solutions with circularity principles having in mind the reuse of materials, resources (WW, BW) and increasing the energy efficiency of buildings by passive and active planning. The uptake of recycled construction materials by other industries boosts **industrial symbiosis at local/regional level.**
cost savings	**Waste prevention, re-use and similar measures could bring net savings of €600 billion, or 8% of annual turnover, for businesses in the EU**[4]. Many companies have observed that linear business models increase their exposure to risks; notably 75% of EU SMEs have seen their material costs rise in past 5 years[5] along with increased supply disruptions. The situation can worsen if no action is taken due to the constant rise in urbanisation and resource extraction costs. On the other hand, the current saving cost of all HOUSEFUL solutions at building level depend on a case-bycase and the current normative at city level. These are first estimations based on a preliminary evaluation: **10% (on average) saving costs** in the long run **of using treated WW and recovering of energy** at building level **with respect of external energy and tap water costs**; **80% (on average) saving costs** of **using treated WW and recovering of energy at building level with respect of all external costs** (including externalised costs such as investment in infrastructure cost); **70% (on average) savings** from current fees applied to citizens **for the treatment costs of WW** and **waste** management of BW.

SOCIAL BENEFITS

Employment	Not only squandering of billions of € of quality materials, many thousands of potential EU jobs are also being thrown away through the "cradle to grave" approach. **At EU level, if a target of 70% for recycling of key materials is met, up to 322,000 direct jobs could be created and an additional 115m tonnes of glass, paper, plastic, ferrous and non-ferrous metals, wood, textiles and BW could be recycled**. These jobs would have knock-on effects in up- and downstream sectors and wider economy, creating 160,900 new indirect jobs and 80,400 induced jobs[iii].
leasing/ renting services	HOUSEFUL will co-create 11 CEBOs for EU markets for the waste, WW, energy and material flows will be designed in such way that renting, leasing, customer services and capacity building services are promoted over ownership. Since the responsibility of the service/product/technology lay on the provider and not on the costumers the quality of the services is higher and much more in demand.[iv]

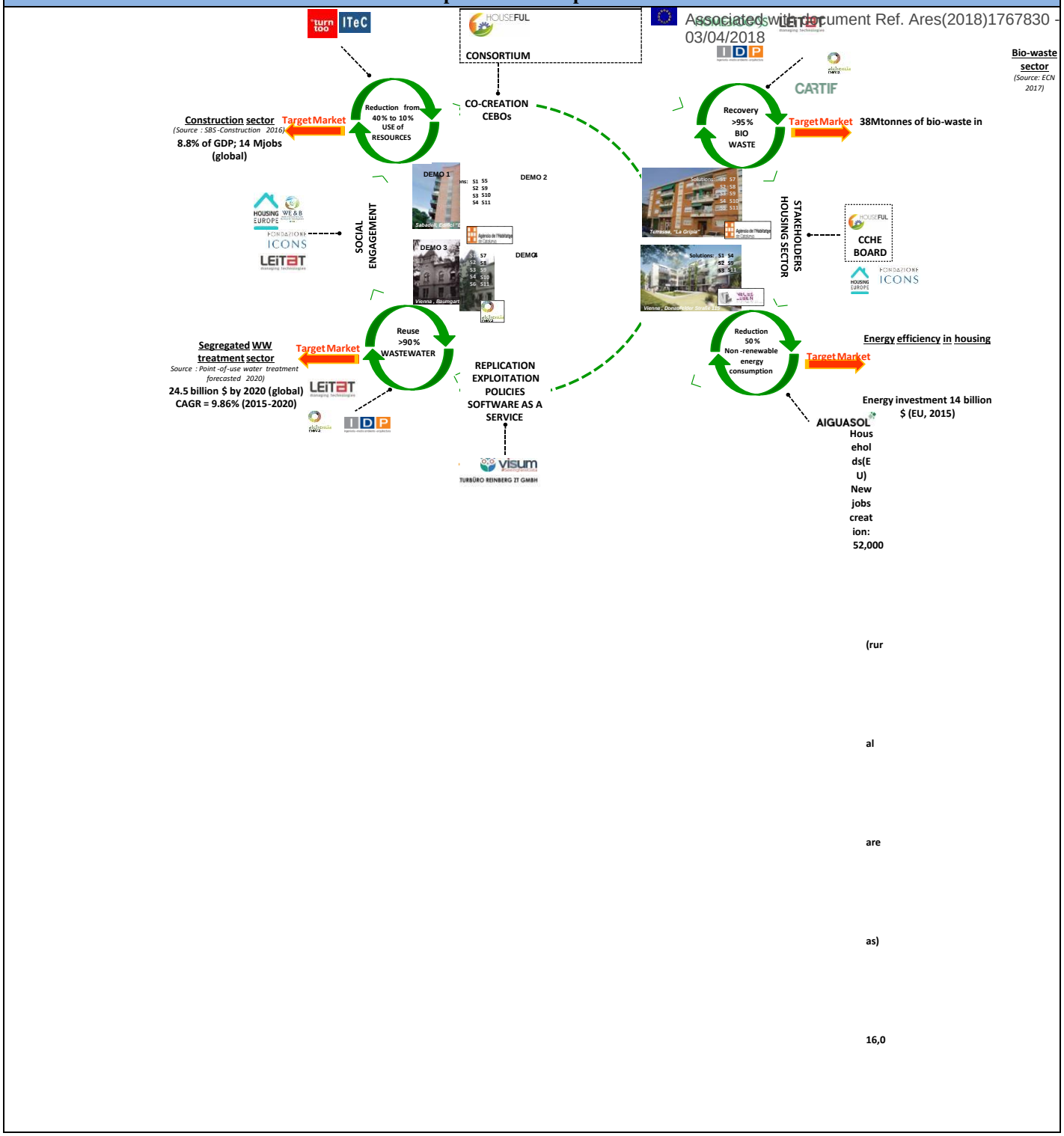
ENVIRONMENTAL BENEFITS

GHG emissions	Housing sector is estimated to cause 35% of GHG emissions[7]. HOUSEFUL will tackle the energy efficiency of existing and newly constructed buildings in two different ways: 1) by using a µCHP to provide end-users with heat and power produced from WW and BW and 2) by planning and designing the use of more energy efficient products and services in the preventive, corrective and replacement phases of selected residential buildings. With these solutions, we estimate a **reduction of up to 60% of CO_2 emissions, contributing to the efforts to limit the global temperature rise to 1.5ºC, as provided for in the COP21 Paris Agreement** (as expected in the **EI 8** of the topic).
resource efficiency	The optimal use of the water, BW and materials cycles within the building will **reduce by up to 40% waste** ending in landfill, allow for the **recovery of up to 95% of food waste** and the **recycling of >90% of WW** (grey and black WW) **for water reuse** (e.g. irrigation, toilets), **improving the energy efficiency** of buildings **of up to 30%**.

hazardous substances	The **40% reduction in wastes disposed in landfill** will significantly reduce the risk of hazardous substance released into the environment (e. bituminous mixtures, solvents, paints) strongly harmful to the environment and human health. The project will also ensure the correct management of wastes/resources likely containing persistent organic pollutants and will follow standards published or under approval by the CEN/TC 351 to allow the selection of sustainable materials

The **outline of the business plan of HOUSEFUL** is summarize in the following table:

Targeted market, size and location
Target market: Social housing market in Europe key figures (2012): 1) 267 000 new dwellings completed**, 2)**155,000 dwellings refurbished, **3)** 38 529 430 000 Euros investment, **4)** 88 480 107 000 Euros turnover, **5)** 7,170 staff employed by the federations, **6)**369,024 staff employed by local housing shows the market status in Spain and Austria (countries where the demo buildings are located)
Expected Market position



00 (citi es) sect or Zeroenergy buildings construction (Source: IEA, 2016)
Stakeholders (St) and their needs
Stakeholders: The main stakeholders of the services developed in HOUSEFUL are local/national authorities owning and/or managing social housing infrastructures, as well as the potential tenants and dwellers The other important stakeholders are identified above (**St1St5**). / Knowledge **needs/concerns**: Identified needs are presented in Table 1 of the excellence section (**N0-N11**). Each need is linked with the corresponding solution (**S1-S11**) and the St implicated. Additionally a participatory process to address concerns and benefits perceived by the stakeholders will be integrated in the decision-making process and the co-creation of services.
Circular Economy Business Opportunities (CEBOs) and value proposition of HOUSEFUL
Business model: Mixed models combining the primary investment in purchasing and installing the technologies and participation to secondary services/products resulting from the "circularity" and shared services approaches. / **Value proposition**: The innovative circular economy approaches and greentech technologies demonstrated by the HOUSEFUL can be integrated as recommendations and/or conditions in the documentation for technical projects of new builds, reconstruction and retrofitting of existing infrastructures.
Market Barriers and Risks (more detail in Table 10)
Legal Barriers: Lack of existing policies at European level for the proper regulation of the implementation of circular economy services at households including the use of treated WW, the production of biogas, and the installation of NBS at building level, may delay the market uptake and acceptance of HOUSEFUL services / **Economical Barriers**: HOUSEFUL services may have difficulties to reach niche market and other markers if there is an insufficient investment plans from public administration and/or the CEBOs are not properly replicated. The high cost of innovation may also hinder the implementation of circular services in social houses / **Social Barriers**: Low interest and acceptance based on unaddressed concerns among homeowners and other stakeholders may affect the uptake of the services, as well as lack of capacity and understanding to implement and/or uptake solutions

EI 4: Facilitate the inclusion of resource or materials criteria in designing products/services (e.g. durability, reparability and recyclability).

HOUSEFUL will promote the reparability, durability and recyclability of the technologies, materials and products by identifying the main materials-elements-products-components at the 4 demo buildings before taking further action. This will permit goal-specific maintenance operations (preventive, corrective or replacement) allowing for the use of more energy/material-efficient products and services, identification of current materials flows (resources used and wastes produced) and the reuse and recycling of construction materials from potential maintenance operation, refurbishment and/or demolition, together with the possibility of reverse logistics at local/regional scale.

EI 5: Create new business opportunities for industry and SMEs in the EU.

In general terms, circular economy solutions, enabled by the technology revolution, would allow Europe to grow resource productivity by up to 3% annually. This would generate a primary-resource benefit of as much as €0.6

trillion per year by 2030 to Europe's economies. In addition, it would trigger €1.2 trillion in non-resource and externality benefits, bringing the annual total benefits to around €1.8 trillion compared with today.

For the housing sector, **a key component of Europe's economic growth (currently responsible for 9% of the EU GDP and the direct employment of over 18 million people)**[v], HOUSEFUL expects to have a high impact on companies working in this sector, which especially characterized by 94% of micro-enterprises (less than nine employees). The implementation of the innovative CEBOs will enhance the circularity of the existing building stocks with the inclusion of new sustainability criteria driven by end-user demands. Accordingly, stakeholders involved in the value chain of the housing sector will have to adapt their current practices to satisfy user requirements. While the housing sector market is currently lacking the information, confidence and capacity to move towards and adopt circular economy solutions, there are already actions going on at EU level to foster the transition and create new business opportunities for European SMEs and industries. An example is the Partnership on Housing and Circular Economy of the Urban Agenda for the EU who has identified cities and their infrastructures as the main engines to carry out the

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transition. In fact, adopting circular economy principles in cities, local SMEs and industries will take part in i) ecodesign concepts, ii) renovation and re-use of sustainable buildings, iii) in industrial symbiosis, iv) the creation of innovative business models for sharing/leasing services, v) valorising urban BW and WW, vi) in circular public procurement. [vi] Other important action is taken by the DG ENV and DG GROWTH developing a common EU Framework of indicators to assess the environmental performance of buildings for all the building professionals. In fact, HOUSEFUL will carefully consider these indicators in the formulation of the circularity level of buildings.

HOUSEFUL will allow SMEs and industries with different profiles to improve and demonstrate innovative solutions at a larger scale and to potentially transfer and replicate all the 11 solutions to other geographical areas or larger industries. Besides this, enterprises will be encouraged to shift their current business models towards a product-as-service approach, relocating investment and modifying income sources placing a renewed demand on housing services such as maintenance, reparation and refurbishment of buildings, further impacting positively on the housing and construction market. The adoption of **circular economy principles by SMEs and industries** will widen their entrepreneurial opportunities and create employment, with the re-investment of saved energy costs expected to create **1M jobs by**

2020, with estimated net saving of €90 billion per year for business by 2020.[vii]

How HOUSEFUL will contribute to deliver innovative solutions to the EU markets of Housing Sector:

HOUSEFUL services will contribute to improve the energy efficiency of buildings of up to 50%, contributing significantly **to save an amount of energy equals to save $ 272,366 M in electricity, gas and water at EU level**.[viii] Passive and zero-net-energy houses are available in many segments of the market but still constitute only a minority of new buildings[ix] . Among the household's related services, housing is the largest direct expense for European households, with an **average annual cost of €9.600, or 27% of direct annual spend** (€15,500, including societal and opportunity costs). Cash-out cost includes purchase price, maintenance, utility costs (energy and water), community development, water treatment, and waste management[ix]. HOUSEFUL aims to avoid the expenses due to externalization by offering an integrated and transversal services, tackling a wide range of issues: reduction by up to 40% waste production, recovery of up to 95% of food waste and its utilization to produce energy for heating, recycling of >90% of WW (grey and black WW) for water reuse. Considering than heating and water treatment are two of the highest sources of expenses in household sector and that the construction the highest consumer of raw materials, HOUSEFUL will deliver innovative solutions to these sectors:

- **Segregated water treatment systems market** (at home level) is estimated to reach **$ 24.5 billion by 2020, with a projected CAGR of 9.86%, from 2015 to 2020**. Demand for water treatment equipment is forecast to grow 5.9% per year to $ 13 billion in 2017 while commercial and residential markets for water treatment are expected to be among the fastest growing markets for equipment worldwide; however, high equipment cost acts as a restraint to the market. Companies and manufacturers have to continually invest in R&D activities for innovative solutions in order to adhere to the standards set by governments for water treatment systems.[x] To answer this challenge, HOUSEFUL offers innovative WW treatments by means of the utilization of two solutions (**S5 & S6**): outdoor and indoor NBS (green façades and vertical ecosystems) and O_3/UV disinfection systems. Such process will be carried out within the buildings, providing fresh and re-utilizable water without supplementary costs related with water transportation or storage. Moreover, the required energy for the generation of O_3/UV is produced indoor, allowing a considerable saving for energy purchase.
- Concerning the **heating**, around 85% of heating and cooling is produced from natural gas, coal and oil products, and only 15% is generated from renewable energy.[xi] Furthermore, as the cost of heating is an important element of household expenditure, which can amount to up to 20% of a household's income, special attention should be paid to measures related to energy efficiency in heating and cooling systems along the entire energy chain. Considering the impact of heating on Europe's final energy demand in the residential sector (65% in 2010), there is

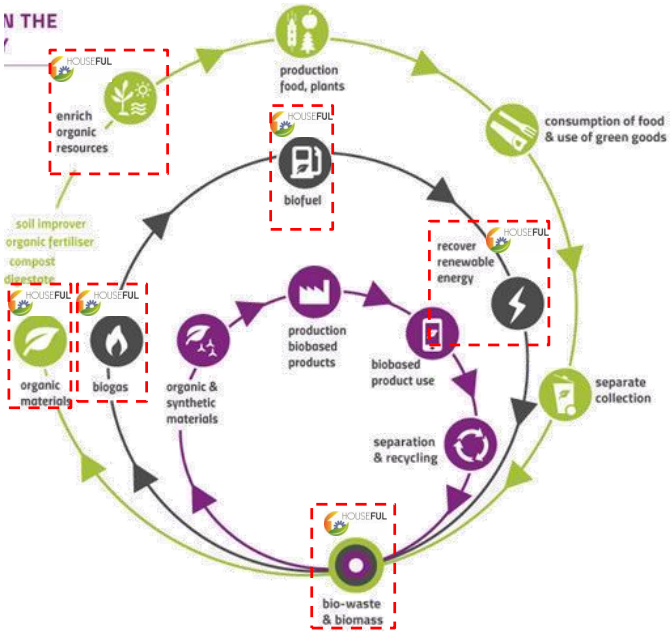
a clear need, but also an important opportunity, to involve households more in this area.[xii] As mentioned, HOUSEFUL services include the production of heating and power (**2600 kWh/y renewable energy from biogas)** from the reutilization of home produced BW and blackwater using AnMBR and dAD for biogas production: such approach will increase the energy efficiency of building up to 30% while reducing the price for energy purchase from external sources. Moreover, the utilization of an interactive tool will make the end-users self-accountable on the utilization/saving of energy in their own houses.

- **Construction materials**: A relevant contribution to improve the heating/cooling services can come from the utilization of new sustainable construction materials. **Construction is one of the largest economic sectors of the European economy, representing 8.8% of GDP and almost 14M jobs**. On average, 10-15% of building material is wasted during construction: 60% of discarded materials is either put in landfill or incinerated, while only 40% are recycled or reused. The lack of utilization of sustainable materials in buildings edification, together with a low level of recycling, generates 25-30% of all waste produced in the EU. In value terms, Europe lose 95% of the material and energy value. On average, Europe uses materials only once generating evitable expenses for materials supply. On the other hand, with a thorough utilization of sustainable materials, 20-40% of energy in existing buildings can be profitably conserved. 99% of enterprises in the European housing sector are SMEs, considered as a key driver for economic growth, innovation, employment and social integration; they produce 80% of the sectors turnover.[xiii] Despite this, SMEs find difficult to break into the market for larger projects. In 2013, 40% of SMEs active in housing sector were failing to win nine out of ten public sector contracts and more than half reported their success rate falling over the last five years. This is partly due to the cost and difficulty of prequalifying and tendering.[xiv] HOUSEFUL will stimulate this market segment by involving the SMEs, as well as all the Associated with document Ref. Ares(2018)1767830 - 03/04/2018 other partner of the consortium, in innovative circular models which will boost the creation of new jobs and a "greener" approach that considers the use of sustainable materials and will contribute up to 40% reduction of wastes disposed in landfill.[xv]
- **Market of BW treatment**: BW accounts 42-60% of the total composition of municipal solid waste. **Europe produces 90 Mt pear year of BW of these, 38 Mt from households)** and is already recycling only 30 Mt. Thus, 60 Mt are disposed to landfill being the actual potential of BW recycling with circular economy principles. In fact, the European Compost Network states that BW collection is technically and economically practicable in any municipality, and that the costs could be kept low if the treatment is optimise at home level. Overall, **the potential of BW sector that is now unprocessed would add 52,000 new jobs in rural areas and 16,000 in urban areas**.[xvi] Figure 6 shows HOUSEFUL contribution with the solutions S7 and S8 to the circular economy biological cycle for BW management.

EI 7 & EI 8: Provide evidence-based knowledge regarding the enabling framework conditions (such as the regulatory or policy framework or cultural factors) that facilitate a broader transition to a circular economy in the EU; implementing SDGs, in particular SDG 12 and COP21 Paris Agreement.

i) <u>Policy framework in line with HOUSEFUL project</u>

HOUSEFUL will facilitate the separate collection and proper treatment of household BW producing environmentally safe compost and biogas, breaking the link between economic growth and the environmental impacts associated with the generation of waste. HOUSEFUL is thus in line with the **Waste Directive 2008/98/EC**[9]. Furthermore, HOUSEFUL will increase the awareness and demand for sustainable buildings, fostering interest in resource and energy efficient practices and technologies and thereby working towards the fulfilment of the **Energy Performance of Building Directive (2010/31/EU** in review) and the **Energy Efficiency Directive (2012/27/EU).** Even though HOUSEFUL is in line with the most important regulatory framework, the consortium has already identified some **social/legal/technical barriers** such as:



a) lack of sufficient information to end-users concerning the importance of an energy saving within the housing activities, b) an insufficient level of separate collection of waste (on average, only 19% of generated municipal waste is collected separately in EU capitals)[10], c) lack of existing policy at EU level on resource use in buildings other than for energy efficiency[11], and d) lack of a common EU directive that fosters the reuse of WW (**Urban WW Directive 91/271EEC** recognises the reuse of treated WW whenever possible, however no EU regulation and

common standards for water reuse exists). In order to unlock the full potential of WW at residential level, the project will consider national/local legislations to choose the best approach to reuse water. HOUSEFUL will overcome these barriers by proposing a model that considers the wide range of EU priorities included in the **EU Action Plan for CE**[12] and **7th Environment Action Programme**[13]**.** The consortium will identify regulatory uncertainties that may hinder the application of solu- **Figure 6. HOUSEFUL solutions S7 (valorization of biowate for** tions and will raise its concerns to the EC via the "**In- biogas production) & S8 (stabilization of BW for compost in the** xvi **novation deals".** The impact of HOUSEFUL in the **biological cycle for circular economy (Source: ECN, 2017).** main directives affecting the water, waste, material and energy vector of the housing sector are included in Table 8. In addition to the regulatory barriers, HOUSEFUL will have to overcome **cultural barriers** related to the willingness of inhabitants to use reclaimed water with confidence, to properly separate BW from other types of household wastes and to install the technologies available for domestic water and BW management cycles. To tackle cultural barriers the project will design strategic dissemination and awareness campaigns with the participation of two public bodies on social housing and the cooperative CECODHAS.

Table 8. Regulatory and Policy Framework affecting HOUSEFUL and main activities of the project to address them

Regulation	How affects HOUSEFUL services	What will HOUSEFUL do	WP
COM/2014/ 044 (Re-source Effi-ciency)	The main objective is to promote a more efficient use of resources consumed by means of new and renovated residential and public buildings and to reduce their overall environmental impacts throughout the full life cycle.	HOUSEFUL aims to minimise waste destined to landfills towards 30%; recovery >95% food waste at home level; recycling >90% of rainwater, greywater and blackwater; to improve the energy consumption of buildings of up to 30%.	WP2 WP3 WP5
COM/2016/0105 and Implementation of Di-rective 91/271/EEC (Urban waste water treatment)	**A)** Member States with high compliance rates usually show such trend all over their regions. However, lower compliance rates are often due to poor performance of some of their regions (i.e. Spain). **B)** Help Member States to achieve the objectives of the UWWTD, a number of actions are taking place, such as bilateral dialogues, developing implementation programmes, and **improving management and dissemination of implementation data.**	**A)** HOUSEFUL will contribute to break w Associated with document Ref. Ares(2018)1767830 - 03/04/2018 such tendency by targeting, thorough a demo site in Spain, the achievement of >90% water reuse at residential level. **B)** The Project dedicates an entire WP (W well as several activities in other WPs to the dissemination and spreading of the services relat the reuse/save of water in residential buildings.	WP2 WP3 WP4 WP5 WP7
Directive 2008/98 (Waste)	Core on prevention, reuse and facilitation the separate collection and proper treatment of BW in order to produce environmentally safe compost and other BW based materials (35).	HOUSEFUL main objective includes mini-mizing the waste destined to landfills towards 30%; recovery of >95% of food waste; the Project offers the generation of renewable energy for own consumption (i.e. CHP) from biogas, and the compost production for gardening purposes for self and neighbours.	WP2 WP3 WP4
Directive 2010/31 (Energy perform-ance of buildings)	**A)** The building sector is expanding, which is bound to increase its energy consumption. **B)** A cost-effective or cost-optimal balance is mandatory to get energy efficiency levels. **C)** Major renovations of existing buildings, regardless of their size, provide an opportunity to take cost-effective measures to enhance energy performance.	**A),B)** HOUSEFUL boost the energy efficiency of buildings of up to 30% and the generation of renewable energy (e.g. utilization of >95% kitchen waste for energy valorisation, convert water and >95% food waste into a renewable source of heat/power and compost). **C)** Project services target new as well as zeroten year old buildings, regardless of their size.	WP2 WP3 WP4 WP5 WP6
COM/2012/ 0433 (Sustain-able com-petitiveness	Energy performance of buildings and resource efficiency in the use of products for the construction of buildings has an important impact on energy, climate change and the environment. Market support is also required for all market actors.	HOUSEFUL services will increase significantly the energy performance of buildings; Project business models and market strategy are meant to exploit the HOUSEFUL services engaging a wide range of different actors.	WP4 WP5 WP6 WP7

ii) Alignment of European Circular Economy Roadmap with HOUSEFUL project

HOUSEFUL is aligned with the global European commitment to create a more sustainable and Circular Economy by shifting the traditional industrial model of housing sector to a systemic eco-innovative model that goes beyond single technologies/services, creates new business models and involves individuals, political and community entities. The project supports the roadmap for a systemic approach to eco-innovation, specifically by f**ostering the social housing sector market, especially the markets for new CEBOs in BW, WW, energy efficiency and building materials.**

Strengthening the adoption of circular principles in the construction of new buildings and in their renovation will be needed, supporting citizens/consumers in the transition to the Circular Economy, and continuing the progress towards a harmonized Life Cycle Assessment dataset.

What HOUSEFUL has to offer in terms of Circular Economy is outstanding. Projects like this creates new circular economy business opportunities for the housing sector that are more innovative and less dependent on the import of materials and non-removable sources, while at the same time reduce adverse effects to the climate change. These initiatives are considered to be acting at a micro-level but it is important to point out that Europe's economy is able to scale these ones up, hence opening the door to an enormous potential for economic growth.

iii) <u>HOUSEFUL project and the SDGs and COP21 Paris Agreement</u>

HOUSEFUL will also contribute to the implementation of the SDGs (**EI. 8**), especially **SDG 12** by using reclaimed water and thus replacing the use of tap water where is not needed (e.g. gardening, flushing toilet, etc.).[xvii] Additionally, SDG 12 claims that residential energy use is the second most rapidly growing energy use (after transport), thus HOUSEFUL will contribute significantly to the promotion of energy efficiency gains with its technological and non-technological advances. Likewise, the promotion of BW valorisation and compost use within buildings (i.e. greenwalls) or gardening purposes for self and neighbours will not only prevent waste generation, but also all waste-related-energy consumption. For the first time ever, the housing sector's critical role to fight against climate change was recognized at COP21 in Paris where it was stated that in order to achieve the global CO2 emissions by 60% in 2025 compared to 2012, the building sector will require a substantial renovation of the existing stock buildings as well as the construction of new buildings with highly energyefficiency, near o net-zero energy consumption. With these measures, the housing sector will avoid at least 50% of the foreseen growth in energy consumption by 2030.[xviii]

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<u>The improvement in innovation capacity and integration of new knowledge of HOUSEFUL consortium (Table 9) is key to achieve the expected impacts:</u> During the proposal preparation phase, the HOUSEFUL consortium has been already strongly committed for the identification of the above identified needs in the housing sector and the subsequent co-creation of services. Such co-creation of services, the core objective of HOUSEFUL project, could only be developed if the project counts with the necessary stakeholders that involve the whole value chain of the housing sector. First of all, the project will tackle the identified needs in the social housing sector by involving **two public agencies for social housing**: The Catalan Agency of Housing (**AHC**) owners of the frontrunner demo buildings in Spain and **Neues Leben** holding the Austrian demo building "Donaufelderstrasse". Both agencies are key partners in the consortium, not only by proving three demos building but also to ensure an effective co-creation and social engagement with the one of the main stakeholders: homeowners, tenants, dwellers as well as with municipalities and other social housing entities. In fact, due to the participation of **CECODHAS**, the CEBOs will be expanded in other countries with the involvement of its network in 24 countries gathering **43,000 housing providers**. All together, they manage over **26 million homes**, which is about **11% of existing dwellings in Europe**. The social analysis and development of effective engagement strategies for an effective co-creation of the services (WP 3) will be driven by the experts of the SME **WE&B** working on social aspects influencing behaviour and choice-making of stakeholders within the environmental sector. The impact on society in terms of awareness, capacity development, and uptake will be also supported by **ICONS**,

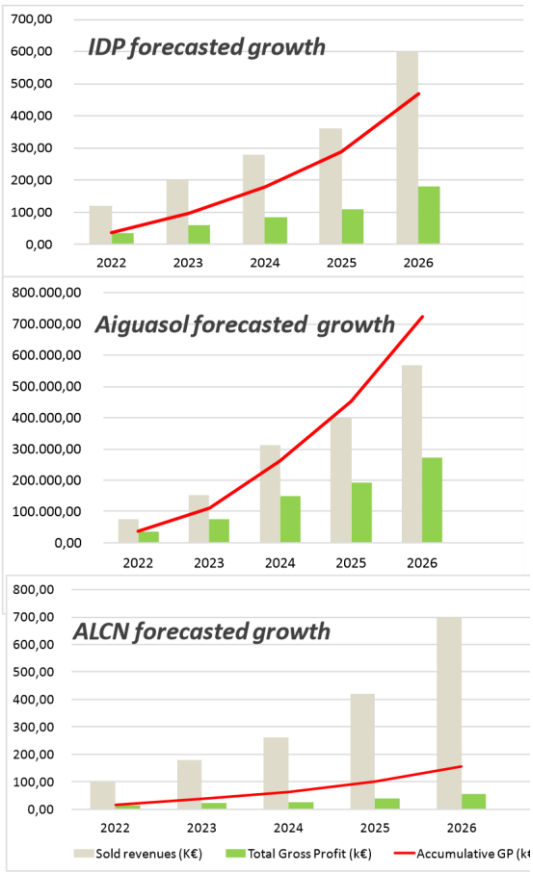

Figure 7. Forecasted growth of the SME IDP involved in the Spanish frontrunner buildings

with experience as leader in communication and dissemination activities in large project in the urban environment. From the technical point of view, the different systemic solutions are tackled by the participation of internationally recognized European SMEs with long trajectory in each of the market sectors of addressed vectors. **Material vector**: **Turntoo** are experts in future buildings and in the creation of material passports, a key decision tool to choose smart materials and safeguard existing ones for boosting circularity within building retrofitting. Turntoo will closely collaborate with the RTO **ITeC,** a technological center working in the construction sector**,** by feeding the material passport with the appropriate BIM model. **Water vector**: Nature-based solutions implemented in the Spanish and Austrian demos for the treatment of segregated WW and rainwater are conducted by **ALCN**, an active SME in Austria and Europe in boosting natural

resources with long-term sustainability and social benefit. ALCN brings to HOUSEFUL the 4th demobuilding "Baumgartenstrasse". ALCN foresees a broad co-financing campaign of the planned €1.700.000 for this demo house which will enable the creation of **5 new jobs with a net profit of 155 k€ by 2026**. **IDP**, an innovative and multidisciplinary SME well positioned in the WW and environmental engineering sectors in Spain and Latin America will be responsible of designing the works in the Spanish demos, both for the water and waste vectors. By participating in this project, **IDP forecasts the creation of 15 new jobs and a ROI of 23% with a net profit of 470 k€ five years after the end of the project** (Figure 7) **Waste Vector**: ALCN will be also involved in the treatment and valorisation of BW and blackwater at home level to produce biogas. For this purpose, two technological centers collaborate with this SME: **CARTIF** and **LEITAT**. **CARTIF** is the leader of the WP 4 of the demos due to the center's wide experience with large scale demonstrations in buildings (see section 4). Moreover, **CARTIF** will perform the cleaning process of the biogas produced in one of the Austrian demos. **LEITAT**, the coordinator of the proposal and the generator of the idea will participate in the overall process of the project guiding the partners towards the successful implementation of all the tasks. **LEITAT** lead the environmental, economic and social assessment as well as the development of a compacted and transportable AnMBR to one of the Spanish demos. The collaboration and results of the project will expand **LEITAT's knowledge portfolio** and support new projects and industrial R&D and innovation services with a total value estimated **at € 1-2 million over the next 5 years**. As a result, our circular economy department will grow by 1-2 employees to support the increased business activity. In addition, **HOMEBIOGAS** will bring their AD innovation to treat food waste into biogas. Thanks to HOUSEFUL the company will **increase the profit by 1.7million € and hire 36 new employees by 2023**. For the **energy vector AIGUASOL** is participating due to their experience on energetic plans (i.e. the Barcelona energy improvement plan), building thermal and lighting simulations, and renovation proposals definition and validation through cost optimal analysis (i.e. the TOBEEM project introducing a semi-automatic cost optimal analysis for tertiary buildings, for more than 20,000 scenarios). **AIGUASOL** foresees **the creation of 7 new jobs and a ROI of 21% with a net profit of 725 k€ five years after the end of the project (Figure 7) GWR** is a social and sustainable architecture consulting company that similar than IDP will work for all the architectural and technical issues that will arise in the Austrian demos. Lastly, the Irish SME **VISUM** will develop the web-based platform to implement the HOUSEFUL SaaS to enable the users to access to the co-created HOUSEFUL services and foster their replication. VISUM foresees the creAssociated with document Ref. Ares(2018)1767830 - 03/04/2018a-
tion of 1 new position 2022 and 7 by 2027. The project will improve the innovation capacity not only by the capacities of the partners of the consortium but also by counting with 31 experts of the whole value chain of housing sectors in the CCHE Board. These experts have expressed their interest with a LOS and include representatives of key experts in the sector of construction, environmental consultants and experts in circular economy. (Annex 1). HOUSEFUL also counts with the support of the following cities: Vienna, Viladecans, Valladolid, Ludwigsburg, Terrassa and Sabadell, which together with the CCHE Board (see section 3.2 and annex 1) will engage at least 10 follower buildings for the replication of HOUSEFUL.

Table 9. Companies previous knowledge and innovation expected increase due to HOUSEFUL

ALCHEMIA **(SME)**	ALCN has broad knowledge on circular economy topics and on NBS. They developed a series of close loops in building as showcases of innovation.	Demonstration of the potential of NBS integration to different plant based technologies in many EU These flagship buildings can set new opportunities for and national projects ALCN to get involved in further building projects.
VISUM **(SME)**	Several projects for software development, Process analytical technology (PAT) &industrial symbiosis	New sector and target application
AIGUASOL **(SME)**	Large experience on energy calculation proce- dures including several own or shared with third parties commercial softwares. AIGUASOL also has developed the design of buildings under some economy principles. developed and user oriented services.	Increase its knowledge related to the energy vector understood as a service not for the designers and end-users of the buildings. Commercialization of circular economy oriented services related to the new SaaS circular consultancy services. Value chain analysis for energy-intensive sectors: Get deep knowledge on social housing market in Europe buildings, industry (process industries) and trans- and selected countries/regions. Understand the needs port (urban mobility). Mapping of business mod- and constraints of the residential infrastructures owners and managers. Share and brainstorm on new business models for innovative service with different European ness model assessment to accelerate the market partners and actors uptake of Energy Efficient Buildings
LGI **(SME)**	Market analysis, stakeholder mapping and busi- ness mod- els for emerging and existing energy systems.	
GWR	GWR works in cutting edge architecture with the	

(SME)	specialisation on passive house buildings. We work daily with innovations in the building sector.	Acquire knowledge on biogas production at households.
TURNTOO (SME)	Designing and implementing new business models to reduce resource dependency.	Creation of Material Passports for social housing sectors across Europe. Reviewing the creation of CEBOs
CECODHAS (Non-profit)	Experience in fostering access to decent and affordable housing. They provide housing for socially sustainable people.	Contribute to the adoption of circular principles in the social housing sector by close contact with their members.
WE&B (SME)	Water and environmental solutions, innovative business practices and builds capacity to assist communities, companies and authorities to sustainably. Social innovation practices to ensure that society patterns and factors conditioning mental issues are included in the development of new products, processes and services to foster market uptake.	To gather more knowledge about social and environmental in circular economy of building sector choices
(Non-profit)	Improvement of human capital through public communication of European research and innovative cultural initiatives and citizens' co-creation activities and stakeholders' feedback. Application increase of statistical data will also implement ICONS **ICONS** commu- baseline of data to benchmark the level of engagement nication and exploitation approach for researchers with aiming at maximizing public and creation strategies of their achievements. Indica- prove ICONS's staff impacts on society in tools to implement its services and uptake. munication and engagement patterns.	Improvement its capacity in terms of knowledge of social assessment through new inputs coming from the tion achievements, ers' feedback. The participation for the benefit of society. of and innovative integrated impact-based our similar projects. The involvement in coand innovators in the project will also further improfessional impacts current expertise and provide tors to measure the research with innovative comterms of awareness, acceptance
IDP (SME)	Strong expertise in both building services and environmental sectors (mainly waste and water). Extensive use of BIM tools during all construction scale (Including Operation & Maintenance 7D) licensed or sold through a spin-off creation.	New solutions to prescribe to the customers. Moreover the development of Biogas AnMBR system at building could end in a new product potentially able to be stages
HOMEBIOGAS	Exploitation of Homebiogas TG1, a system converting household waste by AD into free clean energy	Adaptation of the system to convert higher quantity of BW that the systems currently available.

Factors affecting implementation of Expected Impact and measures to address them Associated with document Ref. Ares(2018)1767830 - 03/04/2018 The following barriers and obstacles that may determine if the impacts will be achieved are identified in Table 10. A proper measure has been thought to counter any such risk.

Table 10. Factors affecting implementation of impact and corrective measures

Barriers/obstacles, framework conditions	Effect on achievement of HOUSEFUL Impacts	HOUSEFUL measures to address factors affecting implementation
Legal Barriers		
CEN/TC 165 – (Waste Water Engineering)	Compliance with the requirements of Standards could trigger delays in the implantation of services	HOUSEFUL Consortium will provide with an update supply of materials and technology in the construction of new buildings as well as in the refurbishment of existing ones
Directive 2010/31 (Energy performance of buildings)	The content of the Directive 2010/31 can be modified so the Project could not fit at best its objective.	HOUSEFUL main objective and services are in line with the content of the EU policies on energy, in particular with the ones of the Nearly zero-energy buildings, which inspires the mentioned Commission' Proposal.
Lack of existing policies at European level on resource use in buildings other than for energy efficiency	Delay in the market uptake and acceptance.	HOUSEFUL services and best-practices will boost the European policies path on resource use in buildings and households. Key local and European representatives will be invited to Project' conferences/workshops.
Need for information on integration within tenders	Slow uptake	Clear advice on how to write tenders which consider circular economy
Social/Cultural Barriers		
Different awareness levels by countries regarding the need of "greener" activities	The diffusion of the services and Project approach can be affected. Reticence of construction industry	Tailored and specific dissemination and engagement activities (e.g. interactive workshops and conferences) will be developed to address all classes of stakeholders involved. (WP3, WP 7)

Insufficient support from the local and public authorities	Inadequate level of the Project' diffusion and exploitation. Limitation on the spread of good practices.	The Consortium includes local authorities and European/local contact point in order to ensure the involvement of public authorities.
Difficulties/obstacles related with the definition of the CCHE board.	Reduction of the expected outcome from the demo-sites. Limitation on Project' sustainability.	HOUSEFUL Consortium and network will ensure an adequate number of highly skilled experts for the composition of the CCHE board.
Low interest and level of engagement of stakeholders at local level	Medium	Continuous monitoring of communication and co-creation enabling to measure their effectiveness. A scalable approach to revise the strategy introducing corrective measures.
Low level of acceptance of end users	Medium	Transparent and easy to understand communication will be developed to convey clear messages about the benefits of the implemented solutions, and concerns and perceived risks will be addressed.
Low dissemination impacts can affect possible future uptake and replication of HOUSEFUL methodology	Low	The multi-target/multi-channel dissemination and communication approach and the collaboration with other initiatives/projects in the circular economy will guarantee the project maximum outreach towards broader audiences establishing a strong baseline for further replication of services
Economical barriers		
Poorly development of BM and business cases.	Difficulties on entering on market and consequent lack of investors. Difficulties/limitations to reach promising segment market (waste and water recycle, sustainable construction materials). Limitation on affordable innovation	The WP4 elaborates engaging business models in order to offer to market an efficient package of services
Insufficient financial support for SMEs to enter in the targeted segment markets.		Cooperation and communication with contact points, clusters and local investors to schedule trade mission. Communication and Dissemination for social and market acceptance (WP7).
Cost limitations for social housing and ↑innovation cost		Cost sharing and demonstration of extra benefits in terms of the Circular Economy

2.2 Measures to maximise impact

Before addressing the planned activities for the efficient implementation of the Dissemination, Communication, Engagement and Exploitation Plan through the WP 3, 6 & 7 and to achieve the impacts described in the previous section, HOUSEFUL partners will carry out three mayor activities with respect dissemination, communication, social engagement, business models and exploitation all we the goal to maximize the impact of HOUSEFUL results and CEBOs:

- **HOUSEFUL Business Modelling** based on the commercialisation of the proposed solutions as a service ("**from solutions to services**") and **market study** and **Exploitation Plan** for the above mentioned sectors (WP 6).
- **HOUSEFUL Software as a service**, SaaS, as an exploitable sharing platform to promote new CEBOs that includes ideas for circular solutions and services for the housing sector and all the stakeholders involved (WP 6)
- Complete strategy for the **co-creation of CEBOS and engagement/activities with the society** Associated with document Ref. Ares(2018)1767830 - 03/04/2018 together with the industry and public regional/national managing social housing programmes.
- The project will be supported by the **CCHE Board** which will not only advise us with their technical/social expertise, but also will help us to engage follower buildings. Section 3.2 presents the first members of the board.

The HOUSEFUL consortium is aware that for the implementation of all the necessary actions and innovative solutions/services in the Frontrunner buildings, **additional co-financing** support not covered by the EC Grant is necessary. This additional co-financing is covered by AHC for Demo sites 1 and 2. For demo site 2, other co-funding will also be investigated. For demo 3 additional co-funding commitments are necessary, however a risk mitigation plan (ref Risk 18) guaranties the implementation of most solutions even if co-funding is not obtained. A Financial and Investment Plan for the frontrunner buildings is provided below:

Financial and Investment Plan for the frontrunner demo buildings

SPANISH DEMO 1 & 2 – Sabadell & Terrassa (Spain) – PARTNER: AHC

Investment Plan:
The refurbishment of demos 1&2 is a priority for AHC due to the extreme high demand and lack of supply of social housing in the area (see letter of commitment in annexes).

Demo 1: AHC has secured an internal budget of 1.5 M€ for the refurbishment of Demo 1, including structural improvements in the envelope (roof and façade), insulation of the façade, solar thermal installation, electric and water installation works, etc. Co-financing would not be necessary for demo 1.
For the investment as such following costs are foreseen:

Investment renovation	Anaerobic MBR	Innovative materials, envelope solutions	Nature-Based Solutions	Commercial O_3/UV	Composting System	Biogas Cleaning
€1.500.000	€ 175.000	€ 200.000	€ 55.000	€ 35.000	€ 18.500	€ 47.000

micro-CHP	Control, instrumentation, SCADA and power systems	Biowaste Grinder	Total
€ 45.000	€ 70.000	€ 20.000	**€ 2.165.500**

These **costs sum up to 2.1 M €**. The financing of this amount should be done in this way:

AHC own funding	EC funding	Total
€1.500.000	€ 665.500	**€ 2.165.500**

Demo 2: AHC is committed to follow circular economy principles co-created by the HOUSEFUL project in demo 2. Renovation works are planned to start in 2020. Even though internal AHC funding is also committed, during this period of time different cofunding opportunities will be investigated. AHC has planned a budget of 1,2 M€ for the refurbishment of demo 2, including mainly envelope interventions.
For the investment as such following costs are foreseen:

Investment renovation	Innovative materials, envelope solutions	Nature-Based Solutions	Commercial O_3/UV	Control, instrumentation, SCADA and power systems	Total
€ 1.200.000	€ 100.000	€ 35.000	€ 35.000	€ 45.000	**€ 1.415.000**

These **costs sum up to 1.4 M €**. The financing of this amount should be done in this way:

AHC own funding	EC-funding	ERDF-funding	Business Agency	Other	Total
€ 1.200.000	€ 215.000	To be defined (M4.5)	To be defined (M4.5)	To be defined (M4.5)	**€ 1.415.000**

Financial resources included in HOUSEFUL: The budget for renovation activities associated with the innovative solutions are not included in the own investment plan of AHC. HOUSEFUL solutions are expected to be financed with the EU contribution for the HOUSEFUL project (1.322.419€), together with the budget associated to other partners (see section 3.4.2):
(1) Water saving measures. New installations, including grey water recycling systems (ALCN and IDP) & greywater and blackwater separation (IDP)***, ***(2)*** Duplicating greywater discharge system for treatment using innovative NBS (ALCN) and commercial O_3/UV higienisation (IDP)***, ***(3)*** Integrating water meters at household and building level: controlling water reuse (IDP)***, ***(4)*** Rainwater pre-treatment with NBS on façade or roof and integration in building (ALCN)***, ***(5)*** Higher standards of envelope refurbishment (e.g. energy efficient facades using modular and more sustainable materials), ***(6)*** Provision of space for separation of BW in kitchens, ***(7)*** Use of materials that contribute to indoor air quality (paint) and recycled raw materials, ***(8)*** Use of materials that include recycled raw materials, ***(9)*** Application of other measures from the SaaS, ***(10)*** Co-creation process with new tenants. ***(11)*** Pay per performance business model: guaranteed energy savings production of PV or Solar Thermal.

***Water measures to be demonstrated and validated in just some flats. The precise number is to be decided and financed by the participant/installer/ supplier (ALCN). AHC will facilitate access for installation and participation of residents and assume maintenance during the lifetime of the project.

DEMO 3 – Centenary Building, Baumgartenstrasse, Vienna (Austria) – PARTNER: ACN

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Investment Plan: For the investment as such following costs are foreseen:

Investment renovation	loft expansion & general costs	winter garden	PV & solar thermal	greenhouse, green facade, compost unit	biogas, burner
€530,000.0	€691,000.0	€228,000.0	€64,000.0	€80,000.0	€110,000.0

These **costs sum up to 1.7 M €**. The financing of this amount should be done in this way

Vienna city funding	EC-funding	ERDF-funding	Business Agency	Crowd funding	Bank loan
€280,000.0	€441,919.01	€482,100.0	€342,600.0	€100,000	€66,381.00

- **Business Agency Vienna, Programme "**Standortinitiative**".** This funding programme offers companies in Vienna to scaleup their production sites. As ALCN is located in the building "Baumgarten" and HOUSEFUL is a large-scale demonstration project for services, business models and derived products, ALCN is eligible for this funding programme and contacted the persons in charge already during proposal preparation. This co-financing of 20% of the overall spending can go up to 500k€
- **Ethical banking crowd funding (GLS Bank, Bank für Gemeinwohl).** GLS Bank and Bank für Gemeinwohl offer crowd funding opportunities to finance projects with ecological and social impact. ALCN is a member of both banks and therefore eligible for submissions. This **crowd funding programme is expected to deliver about 100 k€**.
- Funds within the **Austrian cohesion policy** like **ERDF (S3 Smart Specialisation Strategy, EFRE, etc.)** Will also be addressed. Around € 1.235 billion are allocated for the Austrian regions in the period from 2014-2020. The main challenges, that have been identified for Austria, required for the possible incorporation of these funds, are all fulfilled in this project. They are related to innovation, the labour market, social inclusion and education, and the reduction of CO_2 emissions together with other measures of environmental protection. **The contribution of these funds is usually between 20-50%**. Possible schemes have been identified and as soon as the project receives approval, applications will be incorporated. For the broad participation and co-creation policy of this project, the Austrian ESF will be the right one to address.
- **Bank loan**. If ALCN still needs to co-finance parts of the whole investment plan, the rest will be managed through bank loans. ALCN will first again try to work with ethical banks, EIB through the **ESIF** or **InnovFin** programme and the "Austria Wirtschaftsservice". Ethical banks often give better conditions or even cross-fund projects with high benefits for the common good. So, **this sustainable flagship project could also be a model example of generating common values**.

DEMO 4 – New Building (2017) – Donaufelderstrasse, Vienna (Austria) – PARTNER: NEUES LEBEN

Neues Leben has already carried the construction of this buildings, thus no further investment plan is foreseen. The HOUSEFUL solutions applied to this demo are based on calculations on circularity, BIM Modelling and Material Passport as well as simulations concerning optimisation of energy efficiency and circularity. Main target of the building is to serve as a reference for other buildings, as this building type is equivalent to approximately 70% of all buildings in Vienna. For the building "Donaufelderstrasse".

2.2.1 Dissemination Plan towards defined stakeholders in housing sector and exploitation of results

The introduction to this section summarised the overall approach. To make these activities truly effective they must focus on specific target stakeholders – different communication channels to reach them, different types of information to convey and messages to tailor. In the table below, **main target stakeholders and corresponding dissemination activities** are defined. The preparation of an executive D&C (Dissemination and Communication) plan, in line with the project dissemination and communication strategy will be the first task to be addressed in WP7. It will represent the first milestone governing the project's internal and external dissemination activities. The Plan will identify fundamental elements of the dissemination and communication strategy including: the key messages to be conveyed (WHAT), the tools and channels employed (HOW) and the timing of the planned activities (WHEN), the geographical level (local, European, global) (WHERE) providing a guide for the project's and partners' dissemination activities to maximise the impact of HOUSEFUL results in terms of benefits and acceptance for the end-users (the citizens and the cities as a whole) and replication to support urban regeneration via circular-economy solutions in Europe and beyond. Any effort used and resource invested will be useless, unless the benefits of the project can reach the European society who funded this initiative.

Table 11. Target audience and dissemination activities in HOUSEFUL Dissemination Plan.

Target Stakeholders	Specific dissemination activities

Industry: Construction companies, private promoters, technological and product suppliers, architects, engineers	• Participation in **industrial events and workshops** like: Construmat, iWater, Circular Economy European Summit, European Biogas Conference, Smart Cities World Congress, **European Conference on Sustainable Cities and Towns, , EUSEW, Metropolitan Solutions, CISBAT, BATIMAT, Sustainable Places,** Ecobuild, Future of Building, Bauen und Energie, Smart City Expo, Renexpo, IFAT. • Visibility in **sustainable development information networks**: greenbiz.com, sustainablebusiness.com, businessgreen.com, greenoptions.com, and others. • **Publication of HOUSEFUL CEBOs** for the housing sector: The results from WP6 will be published in relevant industry and technical magazines (e.g. ISWA Waste Management World, Green Builder magazine, Sustainable Architecture). • **Demonstrative sessions of SaaS** will be prepared during the project workshops and events with private companies interested in early testing the service. • **Three technical workshops, co-creation workshops**, and **2 Open House Events and the Final event at European level** will be organised to present HOUSEFUL
	technical and social developments to engage with potential industrial stakeholders.
Municipalities and policy makers, Regional/National Housing agencies. They will be an important target for dissemination to ensure replication of HOUSEFUL services in both in the new design or renovation projects of social buildings. Cities participating in the consortium will be targeted. To ensure the implementation HOUSEFUL, a **policy framework** needs to be tackled. HOUSEFUL will perform specific dissemination measures towards this audience.	• Actively seek interaction with local authorities to engage with them and with society. • Focused publications and communications to National and European public administration associations like: **AVS** (Asociación Española de Gestores Públicos de Vivienda y Suelo), **AEEBC** (The association of European experts in building and construction), • Co-identification of barriers and co-elaboration of ways to overcome them to ease process of admission through the participation in the co-creation workshops. • AHC and NEUES LEUBEN will actively be involved with Catalonian and Viennese municipalities throughout the provision of social housing. Also they will provide **policy inputs** together with CECODHAS **on housing issues** to government ministers and departments responsible for territorial development, housing, and social policy. • AHC has also identified the **Institute of Chartered Surveyors of Barcelona** as an effective dissemination channel as their Sustainable construction agenda and web site is a relevant regional reference point. It has agreed to support the HOUSEFUL.
Academia (Research Institutes and Universities): To ensure continuous research and development of circular services and functionalities in the housing and construction sectors	• Edition of scientific publications in open access **peer-reviewed scientific journals**: Waste Management, Water Research, Bioresource Technology, Energy&Buildings, Open Journal of Energy Efficiency, and/or Energy - The International Journal, CleanTech Republic, Emagazine, The Ecologist, etc • Presentation of results in International scientific conferences (besides those listed for the industrial sector above): Conference on Sustainable Solid Waste Management, UK AD & BIOGAS and WORLD BIOGAS EXPO, ECEEE Summer study.
Related initiatives under the EC H2020 RDI programme and beyond: To create synergies, HOUSEFUL will seek to collaborate as much as possible with other ongoing projects in the same field.	• HOUSEFUL partners will seek synergies with other ongoing projects (e.g. **ENTRANZE**, **ABRACADABRA**, **Next-Building**, **4RinEU**, **GreenINSTRUCT**, **DREEAM, Plug&Harvest**, and **BAMB**. The project will look at future EU and international initiatives in the same field for example with the Partnership on Housing and Circular Economy of the Urban Agenda for the EU, URBACT programme, EIP-SCC, EIP Water and the EU Building Stock Observatory. • Links with other projects under execution including a number of them in which members of this consortium are involved will be established in the form of common workshops and seminars to bring together the various technologies and services for the creation of circular models in buildings
Public and private investors:	HOUSEFUL, through links with pre-identified region/country authorities, will look for complementary or follow-up funding under the **European Structural and Investment Fund (ERDF)**, **European Social Fund (ESF)** and **Cohesion Fund (CF)**, **European Investment Bank** (ELENA, supporting investments in energy efficiency and sustainable transport), **European Investment Fund**, **DNC Venture**, **Innovative financing schemes**: PPP Public and Private Partnership.

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General public: Dwellers/tenants/citizens will be informed and engage for the co-creation of CEBOs	• **Personal**, **face to face contact to inform and engage the tenants** directly involved in the project supported with paper documents, posters and notifications in the buildings as this is the only effective way to inform and actively engage these citizens. Work will also be done with local residents and neighbourhood associations in order to increase awareness and impact of the project locally in order to stimulate local circular economy initiatives. This work will be supported by press releases aimed at local media (press, radio and television) in order to amplify the message and public meetings to support specific dissemination activities organized by the project. All of these activities will be backed up by informative material on the project web site in local languages (Catalan and Spanish) about the project and specific activities. • Partners will reinforce its presence in **professional and social network sites** (i.e. Facebook; Instagram; Linkedin) and proactively participate in the very active forums and blogs involved in environmental issues.

Technical workshops activities and training sessions: HOUSEFUL will organise **4 technical workshops** (Table 12) with the goal to share the knowledge and results generated by partners and selected stakeholders. Additionally, LEITAT will organize at the end of the project a **Final Event** in cooperation with the organizer of the **European Circular Economy Summit** by M54. The target audience are the industry, the academia, municipalise and policy makers researchers, with an estimated 100-200 participants. The Final Event will include presentations, workshops, hands-on training sessions and a poster session. External speakers will also be invited from other ongoing initiatives in the same field. The Final Event will enable the creation of synergies and facilitate the collaboration between partners to obtain more fruitful results. During the project and in connection to the project consortium meetings, **2 training sessions and 4 study tours will be organised.**

Table 12. List of technical workshops

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Partner(s)	Objective/main topic	Type of training	When	Where	Type of participants
ALCN + CARTIF	Integration of nutrient and water cycles in buildings, barriers and benefits	Water & Waste matters	M40	Vienna	Municipality staff, industrial delegates, water utilities, architects, policy makers
ITeC + **Turntoo**	The main objective of this workshop is to determine the Material Passport (identification of all construction materials used), documenting the products and raw materials used in a building.	Material matters	M36	Netherlands	Construction companies, architects, plant operators
AI-GUASOL + IDP	The main objective of this workshop is to determine the synergies among the current energy building regulations and RES directives with the circular economy business models. Defining how and who will implement specific energy efficiency and RES solutions in leasing and renting models.	Energy efficiency in circular building	M30	Spain	Industry, Manufacturers, ESCO companies and investment funds

2.2.2 Exploitation plan and Implementation of results by HOUSEFUL

Several activities will be carried out during the project to prepare the exploitation of the CEBOs, as explained later in this section. First we describe the vision beyond the project, including the initial envisaged business modelling and the activities needed to reach higher TRLs after the project. These activities will be confirmed during and after the project period.

Table 13: Exploitation plan.

Partner	Exploitable Results/Interest	Routes to Exploitation	Potential Users and Sectors	Complementary funding
LEITAT	AnMBR and dAD applied at home level in the housing sector. Environmental performance of solutions in the demo buildings	Licensing, R&D services, new collaborative projects, or other exploitation strategies such as start-ups or spin-offs in case of promising results and IPR potential	Municipalities, big social housing planning	Other collaborative projects

AHC	Assessment of the value of the waste WW and material in houses. Demonstrate the validity of this Catalan Law 21/2006) and also to provide feedback for future legislative action in this area	Further collaborative research Internal product development: Developing, creating and marketing new products/processes as part of our commitment to innovation Internal service creation: Creating and providing new services provision for AHC's clients	Families/people who use social housing services. Policy makers and regional politicians with competence for social housing and building refurbishment	Own funding, Other collaborative projects Activation of local authorities with competence for social services and housing.
LGI	Knowledge on social housing market and the circular economy business models in residential sector.	Potential support and/or partnership for future exploitation actions of technological partners or service providers	Project partners, service providers	in-kind contribution
AIGUASOL	BCM. Improvement of energy efficiency by active and passive solutions – Building physics service. Guarantee the energy saving/ production in buildings - ESG	Exploitation with other partners of BCM as a SaaS exploited. BPS exploitation as a new service adding HOUSEFUL features and focus on other stakeholders. The ESG service expected to extend current portfolio by adding the residential sector.	Construction companies, technological suppliers, Municipalities and housing agencies, Owners, Dwellers and tenants and Policy makers	No extra founding is expected. Additional funding to add market capabilities and introduce new features. Joint ventures to handle financial risks
VISUM	SaaS	To increase feedback and build the early adopters community and attract additional funding, the SaaS will be licensed for free to users. Further developments at project's end.	free - citizens, public bodies pay professionals, BIM sector	Final stages of development will be covered by private fund of investors or further EU projects (i.e Fast Track to Innovation)
ALCN	Proving a real demonstration case - application of innovations in further buildings	Results used for application in further possible building projects. Knowledge gain for strengthening consulting business	Architects, coapplication in tenders, expert consulting for circularity	National co-funding in this area, other financing options
GWR	Providing energy all year for real passive buildings	Know-how gain for passive house construction and application for new bids. Techno	Official building tender processes, lectures	Building investor groups, building developers
NEUES LEBEN	Possible dismantling of building after life-time, innovative solutions	Applying for tenders with this know-how, possibly also with partners from the consortium	Municipalities, big social housing planning	City governments, private investors for buildings
ICONS	Increased knowledge about users' behaviour, co-creation pattern, social acceptance and indicators for measurement. Knowledge on new BM and new areas to communicate to ICONS audience	Background of future collaborative research projects. New items of HOUSEFUL services and concepts further communicated to the media and the general public through to boost the interest on this topic via online, social and TV media	d with document Ref. Ares .8 Other research organizations and collaborative research projects	ICONS will further self-sustain these research activities based on the knowledge generated by the project
HOME-BIOGAS	Offer a product suitable for multi-tenant buildings to pocess organic kitchen waste and convert it to energy for heating and cooking	Collaborating with other companies in setting ISOs standard for small biogas systems. Key role on defining the specifications required by biogas systems. Ensure that the system comply with requirements	Homeowners or block of buildings	Own funding

HOUSEFUL Project Business Model: In order to describe how the results are put forward by the HOUSEFUL project will maximise its impact, preliminary **Business Plan is presented in the section 2.1**, summarising the expected output for exploitation and target end-users. The BM has served as basis for conceptualising the communication and exploitation included in the work plan. Additionally, each partner has already identified their specific exploitation interests covering products/services as described below and previously summarised. These initial business plans will be assessed in detail as part of the project, using data and intelligence obtained from the detailed market and competitive analysis, and stakeholder feedback that will be undertaken in WP7. The sales of HOUSEFUL CEBOs will not only create return on investment for end user but for the all components suppliers and developers in HOUSEFUL consortium and their subcontractors and suppliers thus creating the economic impact and company´s growth for the sector as detailed above.

Partner: WE&B

Planned key exploitable results and key areas of application: Social perception analysis, stakeholder mapping and engagement strategies; Social engagement methodologies; Creation of BM tools to assist companies to bring their new solutions to the market quicker.
Exploitation Team: We envisage developing, together with the consortium partners of WP3 new strategies and methodologies for stakeholder engagement that can be brought to other industries and products, and increase knowledge on social studies about behavioural insights and attitudes towards building circularity. These techniques will assist our company to grow as we extend to industries that would not have been available to us previous to this project.
Market: WE&B has a strong base in Europe, Africa and Latin America. Social engagement and innovative business practices gained from this project will be transferred to the African and Latin American market.
Business Model: WE&B is a consulting company that sells its services to enable governments, companies and projects to better engage with one another and to develop and innovate in business practices. We have a strong customer base and a wide network of clients and partners. Results from this project will be used in our business model of selling consulting hours.
Financial Projection: The projection for selling these services in Latin America would be firstly in Mexico through established clients and markets with a modest projection of 57,600€ in the first year followed by 115,200€ in year 2 and as we expand to also Africa (South Africa for instance as other major market) we would envisage 172,800€ income. By year 5 we would want to sell our experience and services in other countries like, Peru, Chile and Argentina with a projected income that could reach 500 k€.
Commercialization Roadmap: In order to leverage these services we would use existing channels such as projects (AfriAlliance, Run4LIFE, WaterClima LAC, WACA Programme of The World Bank, GGGI Peru, LCBA Mexico etc) and numerous presentations and working and focus groups that we run in Latin America and Africa to market these new services and experience that we would be able to offer our current clients.
Financial Risk Assessment: The Financial risk for selling these services would be nil as mentioned above, we will use existing channels and established clients.
Partner: HOMEBIOGAS
Planned key exploitable results and key areas of application: Affordable and high performance biogas solution tailored to fulfil specific necessities of households.
Exploitation Team: HOMEBIOGAS have been involved in international projects for creating renewable energy from waste, for "off-the-grid" communities in Africa, Asia and Latin America. The company has been active in Israel since 2010 and formally incorporated HBG in February 2012. Based upon the knowhow and ideas that the founders accumulated during the previous years of field trials and scientific research, the team dedicated the last three years to develop and market family sized biogas systems. Revenues for 2016, the first year that we began offering our products for sale, exceeded €400K. HBG employs 16 people and operates out of our 1,000 m^2 site on the campus of Hadassah Neurim in Beit Yanai, Israel. From its inception HBG decided to outsource all manufacturing work and has established long term relationships with large manufacturers who provide all of the product's components to the company for final testing, packaging and shipping.
Market: HOMEBIOGAS sells its products to end users and distributors in over 46 countries (e.g. Australia, Israel, Italy, Spain, Portugal, Sweden, France, Chile, Kenya, Mexico, Nigeria, South Africa, USA).
Business Model: HOMEBIOGAS has clients in over 46 countries (e.g. Australia, Israel, Italy, Spain, Portugal, Sweden, France, Chile, Kenya, Mexico, Nigeria, South Africa, USA).
Financial Projection: Revenues for 2016, the first year that we began offering our products for sale, exceeded €400K. Up to now, our funding has come from investors who provided us with approx. €2.5 million, in addition to a support of a p-proximately €700 thousand through programs originating from the Office of the Chief Scientist in Israel. Our investors include a Florida-based cleantech venture fund and Israeli, American and British individuals.
Partner: VISUM
Planned key exploitable results and key areas of application: Describe briefly Development status, facts and figures that facilitate the assessment of the potential impact, differences from existing competing products/services, the main IPR issues to be clarified in the consortium agreement.
Market: The SaaS developed within HOUSEFUL will cover a current empty niche of the Building design and building information modeling (BIM) software. Of course the aim of the project is not to compete with the computer-aided design (CAD) products used commonly within the architecture and construction industries, but to offer a set of features currently not covered. Currently available products in the BIM market offer indeed tools and libraries specifically targeted for architectural design and construction, including for example mechanical, electrical, and plumbing. On the contrary the HOUSEFUL SaaS will use the information elaborated with this software to provide an estimation of the "circularity" of the building and will offer the possibility to get in contact new solutions to improve the sustainability of a building and co-create new business opportunity by suggesting new circular value chains. The BIM market is expected to grow from USD 3.56 Billion in 2017 to USD 7.64 Billion in 2022, at a CAGR of 16.51% during the forecast period. This projection in addition to the burgeoning growth of the green building market will create an unmet need of a tool able to evaluate the circularity of new designs and or refurbishment interventions.

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Business Model: The business model for the exploitation of the SaaS will be based on licensing the software via monthly subscription. The pricing model will be the so-called Freemium, which means that basic features will be available for free to citizens and public bodies to allow the assisted evaluation of the circularity of a building. To evaluate improvements based of target interventions or to receive suggestions about further circular solutions a payment subscription will have to be activated. The target conversion rate of free to payment account should be around 5% and 8% according to the volume of account created.

Financial Projection: The development status of the SaaS at the end of the project will be an alfa version. Further investment will be needed by private funding or further European funding scheme (such as Fast Track to Innovation).

Commercialisation Roadmap: The first need from the commercial point of view is to widen the market share of the SaaS to a minimum of 1000 accounts. This will be achieved with an aggressive price policy which will give free access to payment accounts advanced functionalities for 12 months. After 12 months, a discount offer will be offered to those confirming their payment account.

Financial Risk Assessment: Typically the SaaS business model drastically reduce the commercialization costs and the limited initial number of accounts will not require large infrastructure. Nevertheless, a lack of subscriptions to the service will represent a serious problem. For this reason, several measures to maximize the visibility of the SaaS have been planned.
In the case of commercial failure the SaaS can be sold to a major BIM player to be included as module in its software.

Partner: ALCN

Planned key exploitable results and key areas of application: IPR of composting system, indoor green wall and green façade for water cleaning; winter garden for air condensation; own ICT and monitoring solutions for water and air quality measurement and automation; demonstration showcase for possible replication in other buildings. No directly comparable products on the market yet. Describe briefly Development status, facts and figures that facilitate the assessment of the potential impact, differences from existing competing products/services, the main IPR issues to be clarified in the consortium agreement.

Exploitation Team: IPR formulation and development of marketing strategies by ALCN; possible partial licensing to spinoff company blue carex phytotechnologies GmbH; also, establishment of own company-intern marketing, consulting and selling department. Partners involved in the exploitation activities providing information on their previous experience and their business strategies, complementing external resources if not in the consortium.

Market: Building developers, building owners, innovation-affin users of a building, people interested in sufficiency, further innovative projects for demonstrations, municipalities. The potential geographical and economic size of the market, market trends, main competitors and competitive advantage need to be addressed.

Business Model: Organising open house days, talking at conferences, spreading the word, building up a network of possible clients, selling services and products, custom building solutions. Marketing strategy, main clients.

Financial Projection: Until the demonstration case is ready, we will not sell any of the services. Afterwards we believe we can start with one to two cases a year and further developing up to ten a year until 2021. The first investment needed for the demo case will be 1,2 Million €, which we will collect with five different funding mechanisms: we expect around € 280,000 from the city of Vienna, around € 480,000 from ERDF funding, around 340,000 from the Vienna Business Agency, around € 100,000 through crowd funding, and the rest of close to € 70,000 through a bank loan with a guarantee from the InnovFin programme. Sales forecasts, investment needed for covering the costs, funding strategies, including internal sources, loans, equity, public support through for example ESIF , national programmes etc., complementary or parallel projects;

Commercialisation Roadmap: The products developed in the project will be ready for the market afterwards. The prototypes, which will show proof of concept, will be tested thoroughly within this project. The barriers for marketing may still exist in some European countries, which is why we need to assure to be in-line with the countries' legislative framework. This will also be part of this project to identify barriers and ways how to overcome them. the steps planned before the product is ready for the market (i.e. proof of concept, prototyping, demonstrations of technological performance and cost effectiveness field trials, pilots, validation and standardisation issues, regulatory requirements, barriers and how to overcome them);

Financial Risk Assessment: The major risk in this project is the co-financing need. We acquired all possible financing schemes, but still it is not sure if we can secure all these fundings. We already have experience with commercial productisation, even without standardisation. Commercial and production risks, early standardization needs.

Partner: AIGUASOL

Planned key exploitable results and key areas of application: The main AIGUASOL's exploitable results are related to the BCM SaaS (HOUSEFUL Solution 1), the Building physics service (BPS) (HOUSEFUL Solution 11) and the Guarantee the eneAssociated with document Ref. Ares(2018)1767830 - 03/04/2018r-
gy saving/ production in buildings – ESG (HOUSEFUL Solution 12). Related to the BCM SaaS there are currently no other equivalent services on the European market. So, it's expected that the potential of the service is more time-dependent on the maturity of the market than on the third parties' competitiveness. To offer the BCM SaaS, specific issues will be fixed in the consortium agreement with some involved HOUSEFUL partner's mainly related to the circularity methodologies developed rather than previous specific knowledge. Related to the Building physics service (BPS), currently there exists several market energy consultancy services focused to the building sector, Nevertheless, and because of the current market status of the circular economy, only a few companies offer particular circularity services, and usually focused on materials use. It is not expected specific IPR joint issues among partners; further on the one's established for the other related issues as the BCM. Related to the energy saving/ production guaranties (ESG), currently some ESCO companies are trying to offer services like this one, but mainly focused in tertiary sector and/or only for energy generation. It is expected that the new ESG service will

be part of a strategy to be joined with third financial parties once the basics are established within HOUSEFUL, also concerning to IPR issues.
Exploitation Team: Related to the BCM SaaS, the involved team will be the main service developers joined in WP2; specifically, they'll be involved AIGUASOL, LEITAT, CARTIF, ITeC, TURNTOO and WE&B. it's planned that for this service, some of the partner's (AIGUASOL, ITeC) will act as front officers for the SaaS commercialization due to its previous experience, while the rest will participate in the back office tasks (the service itself). Related to the BPS service this will be specifically offered by the energy consultancy partner's (mainly AIGUASOL), even so specific collaborations are expected from other oriented partner's (as ITeC, or TURNTOO) and specific collaborations with third parties in a future commercialization. Related to the ESG service, it's expected to be exploited by AIGUASOL through its spinoff exsergy. Nevertheless, during the HOUSEFUL project developments, some third parties will be contacted and encouraged to join the service exploitation, mainly financial and ESCO companies.
Market: The market for the new brand BCM SaaS, the BPS and the ESG services, is almost the major part of the stakeholders of the building residential sector. That is, the (St1) Construction companies, private promoters, technological and product suppliers, the (St2) Design teams, the (St3) Municipalities and regional/national housing agencies, the (St4) Owners, Dwellers and tenants, and the (St5) Policy makers. For the BCM SaaS, and considering the current inexistence of equivalent products or services at EU level, the potential of the BCM SaaS is quite high and will be more dependent on the time requirements to positioning the circular issues in the market sector. Related to the BPS service, the competitive advantage related to the existing energy consultancy service will be the capacity to offer an integral service and to be well positioned for the new brand market related to the building circular economy procedures and methodologies. For the ESG service, there exist currently several ESCO companies offering equivalent services but mainly to the tertiary sector or related to energy generation. This is considered as a positive fact when launching this new service oriented to residential sector, because it will facilitate the commercial activities and, at the same time, the capacity to sign joint ventures with third parties.
Business Model: The BCM will be exploited as a SaaS service. Some basic parts will be offered for free, while the main core of the service will be implemented probably as a pay per use business model. The BPS will be offered as a new service of the existing portfolio services of the mentioned involved partner's. That will make easier the communication and market strategy to be implemented as the channels and the oriented actors will be the same that currently are on the focus of the related partners. For the ESG, the service will be offered through existent commercialization channels and activities currently commercialized by the involved partners. Nevertheless, it's expected to explore the joint ventures of financial third parties during the project and the exploitation by using these companies' commercialization channels and clients.
Financial Projection: For the initial version of the BCM services, no extra founding is expected. Nevertheless, for further versions of the service, additional funding will be probably required to add market capabilities and introduce new features linked to the building design processes as BIM connection or other relevant characteristics. In this case, also joint ventures with other related companies (BIM providers or so on) will be studied. For the BPS service, it is not expected extra funding to actively commercialize it. For the ESG service, once the new service is implemented in the exsergy spinoff or other involved partners portfolios, it's expected to explore possible joint ventures with third parties (preferably investment funds and ESCO companies) to be able to handle the financial risks associated to the related activities.
Commercialization Roadmap: At the end of HOUSEFUL project the BCM SaaS will be ready to be launched to the market as a beta version. During a reasonable time period of 6 months, the service will be available to reduced pay per use cost and by introducing special offers. From the gained experience, a commercialization campaign will be launched to the main focused clients through the current existing channels of the front office commercialization partner's. For the BPS service, the involved partner's will be able to directly commercialize the new brand service directly through theirs existing channels and by using their regular strategies. Related to the ESG service, during the HOUSEFUL project, and considering the possible joint ventures with interested third parties, a commercialization strategy will be developed. This is because the main expected risks will be related to the financial issues rather than the technological one's and, because that, the commercialization road will be adapted to the joint ventures' models.
Financial Risk Assessment: The financial risks in the first steps of the BCM SaaS commercialization is considered quite low because of the type of service and the capacity of the front office companies to add a service like this one in its current portfolio. Further developments for new versions of the service will be joined with third parties in order to minimize the potential service risks. For the BPS service, not relevant financial risks are expected by the own nature of the service and the experience of the involved partner's in offering equivalent services. For the ESG service, the financial risks could be quite high because of the leasing or renting models to be used in. because of that, the joint venture with financial third parties will be crucial by considering their specialization in the management of suck risks.

Technology Market Transfer: SMEs have identified targeted markets and products to transfer as summarised below:

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Table 14. Technology Market Transfer

Partner	Product/Processes/ Innovation/ Services	Target Market/ Sector	Measures to deliver innovation to Market (during Project)	Measures to deliver innovation to Market (after Project)	Global EU Market Size (M€)	Provisional Sales Forecast
VISUM	SaaS	BIM market		Participation to events and freemium business model	USD 3.56 Billion in 2017	□□□□ version 50 - 100 acc; □ version 200-400 First 2 commercial years 1000
WE&B	Social Innovation and Business Innovation	Water and Environment sector	Social and environmental economics methods	Engagement practices	-	-
ALCN	NBS in buildings for water and air treatment	Building developers	Interested parties	Sold systems in three years after project	7,7 billion worldwide	After2021, 100.000 € revenue
ALCN	Compost cultivator	Composters, home owners	Interested parties	Sold systems in three years after project	~ 40 M€ in Europe	After 2021 35.000 € revenue
GWR	Full energy sufficiency for passive buildings	Tenders and competitions for buildings	Application in tenders with developed innova-tions	New projects received due to this innovation	~ 1,6 billion € in Europe	After 2021 new projects worth 2,5 M€
AI-GUASOL	Building Circularity Methodology (BCM) SaaS	(St1) Construction companies, technological and product suppli-ers, (St2) Design teams, (St3) Municipalities, housing agencies, and (St5) Policy makers	The main innovation transfer during the project will be done through the focussed dissemination activities and related workshops with oriented stakeholders	Measure related to the SaaS offering services, common market dis-semination activities as mailing communications and workshops.	-	In 2026, 725 k€ of revenue
	Building physics service (BPS)	(St2) Design teams, the (St3) Municipalities and regional/national housing agencies, and the (St5) Policy makers	Main innovation transfer during the project through dissemination activities and related workshops with oriented stakeholders	Transferability to 3rd parties ensured through the web explanation of real cases and the publication on scientific reviews.	-	
HOME BIOGAS	TGx Biogas System for European multi-tenant households	Urban and suburban communities	Providing field tests and demonstration site	Develop channels and proactively market our solution	-	-

Research Data Management: HOUSEFUL participates in the Pilot Action on Open Access to Research Data (Horizon 2020 program). The goal is to provide, accurate and high-quality data to the research community so that the project will contribute to future advancements in the field of Circular Economy in the housing sector and the industrial sectors addressed. Table 15 summarises the Research Data Management where generated data and how it will be accessible (if possible) and preserved is presented. Commercially sensitive data are declared confidential and will be protected in accordance with the strategy for IP management.

Table 15. Research Data Management Plan.

Partner	Project Data Generated (WP)	Standards (if applicable)	Routes of Exploitation and data accessibility	Data curation and preservation procedure
All partners involved in WP4	Information regarding the optimization of water, waste & energy vector in the buildings	Presentations: "ppt" or "pptx" Tables: "xls" or "xlsx" format	Research data will be deposited in the free repository Zenodo	Data will be stored on a separated server with strictly controlled access.
ITEC & TURNTOO	Data created for the Material Passport	BIM	Confidential, accessible to demo partners	The data will be made anonymous prior to any use by the project.
VISUM	Data for the creation of SaaS functionalities	Not applicable	Data repository for final data (confidential)	Data stored on a separated server with controlled access
ALCN	Data through sensor data loggers in WP4	.txt, .csv, .xlsx	Data repository for final data	Data stored online at own cloud servers and at a repository for final data
GWR	Drawings in WP4	.cad, .xlsx, docx	Confidential, accessible to demo partners	Data stored in own intranet at NAS
AHC	Personal identification data of tenants (WP 3-4). Energy and water consumption data	These are protected by Spanish data protection laws	<u>Information of tenants</u>: No need for access to this data and it will be protected. <u>Energy & Water consumption</u>: Available for use by other participants subject to the consent of tenants	document Ref. Ares(2018)176783 03/04/20 The data will be made anonymous prior to any use by the project.
WE&B	Social Studies (WP3)	Social Network Analysis (KUMU)	Public	Data will be public through a visualization tool. Other social studies will be published through reports/articles
ICONS	Web and social media statistics (WP7)	none	All the data will be accessible through the public deliverables of WP7	All the data will be included in the public deliverables that will be accessible to all via the public website
AIGUASOL	Building Circularity Methodology (BCM) Building physics service (BPS) Energy saving/ production guarantee (ESG) Energy saving/ production guarantee (ESG)	Several standards and currently existing methodologies listed in WP2 and WP4	The main data concerning to the overall methodology development will be available through the open deliverables and scientific communications.	SaaS implementation of the BCM methodology as a commercial service protected as a commercial software. For ESG, no specific IPR protection will be considered.
NEUES LEBEN	Data exchange with partners	.docx, .xlsx	Confidential	Data stored in own intranet servers
Homebiogas	Statistics on the amount of waste disposed, biogas, consumption of other resources (i.e water, electricity)	ISOs standard for small biogas systems	Research data will be deposited in the free repository Zenodo	Data stored in own intranet servers

<u>Data Management Plan (DMP):</u>

Each work package within HOUSEFUL is likely to generate significant data that will be managed according to the Data Management Plan. DMP aims at ensuring that the digital research data generated in the action will be made findable, accessible, interoperable and reusable in accordance with article 29.3 from the GA Option 1 for actions participating in the open Research Data Pilot, without changing the obligation to protect results, confidentiality, security and protection to personal data (Articles 27, 36, 37 and 39). IPR will be taken into account to ensure that making data accessible does not go against the interests of any beneficiary with regards of protection of intellectual property.

HOUSEFUL project will always obtain informed consent from persons providing data which participation will be entirely voluntary. All activities procedures will be designed according to Directive 95/46/EC of the European Parliament and of the Council of 24 October 1995 on the protection of individuals with regard to the processing of personal data and on the free movement of such data. Within the context of the HOUSEFUL project, procedures for

personal data must be processed in accordance with national and EU legislation, limiting the impact of the persons involved whilst ensuring data quality and confidentiality. Moreover, the DMP will include detailed information on the procedures for data collection, storage, protection, retention, and destruction.
The corresponding deliverable will be issued by M6.and will be updated as new data sets are produced during the project life.

Management of Intellectual Property in HOUSEFUL

Intellectual property (IP) generated within the project will be protected by patents, if appropriate, and its management will be regulated in compliance with the consortium agreement and the IP agreement among the partners. The **consortium has identified three levels of IP** which will be created in the course of the project: i) **Individual and joint IP**, which belongs to individual partners or is jointly owned by partners working in a particular task and is restricted to those partners. ii) **Generic IP**, which can be used by all partners of the consortium. Some of generic IP will be made more widely available to European academics, SME's and other industrial organisations. Iii) **Publicly available IP** which will be published at conferences, on the public web site and made available with no restrictions. The general principles for Intellectual Property Aspects set out by the EC for Horizon2020 projects will apply in HOUSEFUL. Existing know how (background or preexisting intellectual property) of a specific partner shall be made available within the consortium, but strictly limited for use to the achievement of the project goals and for the duration of the project. Each partner owns the foreground he develops, and is responsible for securing IP of the knowledge or results created during the project. **Results:** owned by one or more of the partners shall be licensed to other partners of the consortium on favourable conditions to the extent necessary to enable these partners to exploit their own results. **Publication of results (**Table 16**)**: While project results should be published in the usual scientific form, all concept publications must be submitted to all partners together with a request for permission to publish. An agreement is considered granted if no objection is raised within 1 month after manuscript submission.

Strategy for Knowledge Management and publications: Associated with document Ref. Ares(2018)1767830 - 03/04/2018

Implementation of procedures for assuring adequate protection of potential exploitable results is critical. The partners will be instructed to create routines to critically evaluate the potential for patenting for all the technologies they develop (not necessarily the end products themselves), based on criteria such as innovation, novelty and industrial applicability. The HOUSEFUL CA will also outline the rules for **publication issues**, with the purpose of minimising related conflicts during the project. As part of the HOUSEFUL dissemination plan, the consortium will set up a specific strategy for publication of results. The HOUSEFUL coordination team will have a role in the follow-up of the publication approval process by the management team in distributing the publications and following up the approval process. The MC will be involved by screening the publication drafts. All partners will be responsible for publishing project results in local and international press (press releases or newsletters in magazines and newspapers, etc.) and in peer-reviewed scientific journals and conferences (Table 16). The D&C manager (**ICONS**) will maintain an overview of all published results and in cooperation with the Project Coordination. LEITAT and ICONS, will intervene in case of results which have been classified as publishable do not receive the necessary dissemination. In addition, the dissemination and communication manager will ensure that all project related publications can be read online, downloaded and printed. However, as any additional rights such as the right to copy, distribute, search, link, crawl, and mine increase the utility of the accessible publication, efforts will be made to provide for as many of them as possible. In particular, several open access schemes will be proposed for use to all consortium members. Depending on the nature, the importance, the cost and the available budget, a decision will be taken by the coordinator in cooperation with the dissemination and communication manager and the proposing partner as to the best open access scheme which is to be followed. A dedicated budget for **open access scientific publications** has been reserved, which will be used for **"green" or "gold"** publications in reputable journals. Alternative repositories such as the Open Access Infrastructure for Research in Europe (OpenAIRE), the Registry of Open Access Repositories (ROAR) and the Directory of Open Access Repositories (OpenDOAR) etc will be also exploited, as there are supported by the EC and there is no cost related to using these repositories. In all cases, as soon as possible and at the latest within six months after the publication is performed, **ICONS** will be responsible to collect a machine-readable electronic copy of the published manuscript version in order to be deposited in the project website either, as a reproduction, if permitted by the publisher, or as a direct link to the publication. In addition, the research data and bibliographic metadata needed to validate the results presented in the deposited scientific publications will be stored in the respective area of the repository. Finally, **ICONS** will contact and exploit EC portals and tools (OpenAIRE, Horizon the EU Research and Innovation Magazine, research*eu results magazine, research*eu focus, etc.) to make the HOUSEFUL generated knowledge publically available. Since the tools provide free of charge, online access to the respective publications for any user, they will ensure open access of these publications. **Table 16. Expected publications in HOUSEFUL**

Partner	Publishable results	Possible journals (Expected publications)
LEITAT	BW treated by dAD and AnMBR	Bioresource technology (1); Waste management (1)
CARTIF	Biogas treatment	Waste Management (1)

AHC	Use of innovative materials, products, installations and constructive solutions	*Agenda de la construcción sostenible (4)*
	Achievement of local project milestones and achievements	Local media (2)
	Circular Economy achievements	Collaboration with the UNESCO Chair on Life Cycle and Climate Change at the Universitat Pompeu Fabra. (ESCI-UPF) (2)
AIGUASOL	nZEB building design in refurbishment and retrofitting actions under circular economy principles. SaaS for the calculation and the increment of the circularity of buildings in refurbishment and retrofitting actions	Energy&Buildings, Open Journal of Energy Efficiency, and/or Energy - The International Journal (2) Circular economy European Summit proceedings (1)
ICONS	Journalistic articles	youris.com platform (8 journalistic articles)
ALCN	Cleaning properties of NBS for water and air; compost generation	Science of the total environment; magazines like ISWA Waste Management World, Recycling Magazine, Green Building (6 publications, 1 in scientific journal, 3 in magazines, 2 in newslet- ters/blogs)
GWR	Full energy sufficiency for passive house construction	Sustainable Architecture, Energie & Wohnen (2 publications, 1 in magazine, 1 in newsletter)
NEUES LEBEN	Demolition of buildings with circular principles	Internal newsfeed (1 publication for internal information)
WE&B	Factors influencing behavioural intentions of stakeholders in circularity building	Public understanding of science, newsletters

2.2.3 Communication Plan

While Dissemination is principally aimed at peers, researchers, technical experts and potential adopters and replicators, Communication activities will promote the HOUSEFUL project and its results to a non-specialist' audiences, including the media and the public at large and covering a wide range of audiences, at different geographical levels with the objective of increasing awareness and understanding. To this end, HOUSEFUL communication activities will focus on a few key headlines outlining the project's results, targeting a less specialised audience. Effective communication will be pursued through the project public communication channels and tools at the general public. HOUSEFUL has already identified the following communications activities divided in two groups: **A: Communication materials for co-creation activities with stakeholders in Frontrunner buildings**, **B: Communication materials for awareness** raising activities, C: Public Communication materials to distributed public HOUSEFUL results towards wider audience (Table 17).

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Table 17. Communication activities

Communication Materials		Objective	Timescale	KPIs
A	**call-to-action** video	Support citizen engagement	M8	>100 visits per month
	Simple HOUSEFUL Flyers	Engagement for the interactive calculation tool integrated in SaaS	M8	Distributions >1200
	Social media channels, (Facebook, Linkedin, twitter)	Foster dialogue, enhance public understanding, engagement and acceptance of local stakeholders' and citizens	M1-M54	>500 visits per month
B	**Visual identity**: project logo, graphic elements and templates	Inform relevant target groups about the project, its objectives and expected impacts. Promotion of HOUSEFUL towards different stakeholders	M2	>500 visits per month
	HOUSEFUL Leaflet		M10	Distribution of > 1200
	posters, roll-ups		M1-M54	Presentation in >50 events
	project presentation video	Present the project, objectives and the circular economy concepts in an easy-to-understand way for online distribution via website, social media, communication portals & platforms	M8	>100 visits per month
	HOUSEFUL website	Main online communication channel and interface towards different target audiences. Direct link to SaaS developed in WP6 for stakeholders and citizens will be established	M5	> 800 visits per month
C	**Articles and interviews**	**8 independent articles and interviews** around project-related topics written by professional journalists	M6-M54	Min. 8 articles in total, outreach of thousands online readers reached via project channels and the media

Video News Release (VNRs)	1 VNR will be produced and distributed to European and Word TV	M50	At least 7 take-ups from TV stations, thousands of TV watchers
short web video	HOUSEFUL demonstration outcomes, through simple and direct messages, facilitating the information transfer of more complex contents to a wide audience	M48	> 200 viewers
Press and News Releases	Promotion of project event (workshops, study tours, Open House Event, Final event)	M1-M54	Outreach of hundreds of readers
Pageflow	Digital storytelling tool illustrating and transforming the HOUSEFUL outcomes into marketable solutions	M1-M54	60% of inhabitants of targeted cities

3. IMPLEMENTATION

3.1 Work plan — Work packages, deliverables

3.1.1 Overall work plan structure

The project work plan is structured in different types of actions: i) transversal and management activities (WP1 and WP7); ii) technical and social related activities (WP2, WP3, WP4, WP5) and iii) market and exploitation related activities (WP6). A graphical presentation of the WPs and their inter-relationships is shown in Figure 8. **Project Management** activities are all gathered in **WP1**, which has been conceived to follow up HOUSEFUL performance, thus interacting with all WPs in order to solve the occasional problems and achieve the smooth running of the project.

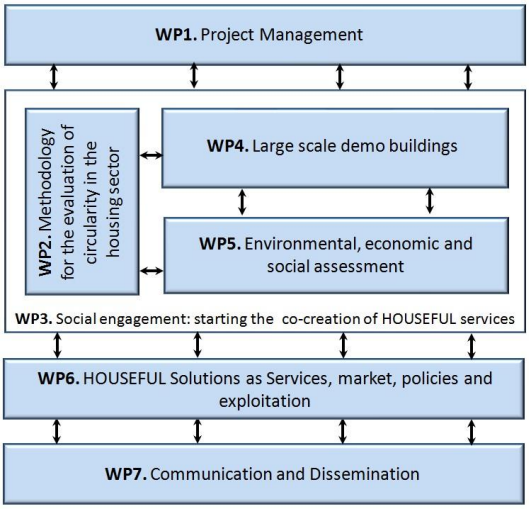

Figure 8. Pert diagram of HOUSEFUL

A **methodology to quantify the circularity level** of the housing sector will be developed in **WP2**, based on calculation procedures and KPIs defined to measure the environmental, economic and social performance during the whole life cycle of buildings. The methodology obtained in WP2 will be used to assess the environmental and economic performance of the HOUSEFUL solutions in **WP5**, and after its validation will be integrated in the SaaS developed in **WP6** to be offered to stakeholders as a circular service. A **social engagement** strategy will be carried out in **WP3** to involve local stakeholders from Frontrunner buildings in the co-creation of CEBOs. The local stakeholders engaged will participate in the co-creation workshops, with the primary aim of gathering the users' needs, identifying potential barriers for the co-creation of solutions as well as raising awareness about the potential benefits of implementing circular solutions. Societal concerns and knowledge requirements will be addressed in WP3 to identify in advance potential risks affecting the implementation of HOUSEFUL solutions in demo buildings (WP4). The HOUSEFUL service integrated by different circular solutions (technological and non-technological) will be demonstrated in four demo buildings during **WP4, Large scale demo buildings.** The data obtained during demonstration activities (WP4) will be used in **WP5** for **assessing the environmental and economic impact** of the HOUSEFUL solutions by means of LCA and LCC. In parallel, the outputs from WP3 will be used in **WP5 to assess the social impact** of the HOUSEFUL solutions.

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The methodology developed in WP2 will be refined during the project duration from the inputs received from WP3, WP4 and WP5. All WPs from 2 to 5, as well as dissemination and communication activities carried out in WP7 are oriented at a successful **exploitation and market uptake** of the project outputs in **WP6**. A market entry strategy of the most promising HOUSEFUL solutions will be developed in WP6 to facilitate the transition from a demonstration phase to a commercialisation phase. The HOUSEFUL service to be exploited will be integrated in a **SaaS developed in WP6**, which will encourage the shift from linear to circular business models. The SaaS developed will include a Circularity Tool based on the methodology defined in WP2 and an interactive sharing platform to exchange successful circular solutions and enhance the replication of proposed HOUSEFUL circular solutions as innovative services for housing sector. Success stories and lessons learned along the different WPs, including policy recommendations provided in WP6 will be transmitted in **WP7 Communication and Dissemination,** which will help to speed up the transition towards the circular economy in the housing sector.

3.1